

# **Mathematics**

Basic 7

Complete BECE Preparation  
Examination Papers & Marking Schemes

**Beacon Educational Consult**

2026 Edition

© 2026 Beacon Educational Consult  
All rights reserved.

No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior written permission from the publisher.

Published by Beacon Educational Consult  
Accra, Ghana

First Edition: 2026

ISBN: [To be assigned]

Printed in Ghana

This book has been prepared in accordance with the  
Ghana Education Service Standards-Based Curriculum (2019)  
and the National Council for Curriculum and Assessment (NaCCA) guidelines.

The publisher has made every effort to ensure the accuracy of the information contained in this book. However, the publisher cannot accept responsibility for any errors or omissions or for any consequences arising from the use of this book.

For permissions and inquiries:  
Beacon Educational Consult  
Email: [info@beaconeducational.com](mailto:info@beaconeducational.com)  
Website: [www.beaconeducational.com](http://www.beaconeducational.com)

# Preface

This comprehensive Mathematics textbook for Basic 7 has been carefully prepared to support Junior High School students in their preparation for the Basic Education Certificate Examination (BECE).

The book contains:

- Multiple choice questions with complete answer keys
- Theory questions with detailed marking schemes
- Curriculum-aligned content following the 2019 Standards-Based Curriculum
- Practice papers from BECE Mock examinations

Each question has been verified for accuracy and aligned with the appropriate curriculum objectives. The marking schemes provide detailed explanations to help students understand the reasoning behind each answer.

We hope this book will be a valuable resource in your academic journey.

— Beacon Educational Consult

Accra, Ghana

2026

BEACON EDUCATIONAL SERIES

---

# Mathematics for Basic 7

A Comprehensive Question Bank  
with Marking Schemes

Curriculum-Aligned Practice Papers  
for Basic 7 (JHS 1) Students and Teachers

---

Beacon Educational Consult  
2026 Edition

Mathematics for Basic 7  
First Edition, 2026

Copyright © 2026 Beacon Educational Consult

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior written permission from the publisher.

This book is designed to support the Ghana Education Service (GES) curriculum for Junior High School. It is intended as a supplementary resource for teachers and students.

The questions in this book are original practice materials created for educational purposes. They are not official BECE or GES examination questions.

Published by: Beacon Educational Consult

ISBN: [To be assigned]

Printed in Ghana

# PREFACE

This book has been developed to support Junior High School Basic 7 (JHS 1) students and teachers in their study of Mathematics, as prescribed by the Ghana Education Service (GES) and the National Council for Curriculum and Assessment (NaCCA).

The book contains a comprehensive collection of practice questions organised according to the curriculum's strand and sub-strand structure. Each paper includes both multiple-choice questions (Section A) and structured theory questions (Section B), reflecting the format of BECE examinations.

## Key Features

- Curriculum-Aligned Content — All questions mapped to specific content standards.
- Comprehensive Coverage — Topic-based papers covering all major strands.
- Detailed Marking Schemes — Complete answer keys with method marks and marking guidance.
- Professional Diagrams — Clear, scientifically accurate illustrations.
- Cognitive Demand — Questions spanning all levels of Bloom's taxonomy.

# CONTENTS

|                |            |
|----------------|------------|
| <b>Preface</b> | <b>iii</b> |
|----------------|------------|

## CHAPTER MOCK EXAMINATIONS

|            |  |
|------------|--|
| Ch 1: Ch10 |  |
| Ch 2: Ch11 |  |
| Ch 3: Ch12 |  |
| Ch 4: Ch13 |  |
| Ch 5: Ch14 |  |
| Ch 6: Ch15 |  |
| Ch 7: Ch16 |  |
| Ch 8: Ch17 |  |
| Ch 9: Ch1  |  |
| Ch 10: Ch2 |  |
| Ch 11: Ch3 |  |
| Ch 12: Ch4 |  |
| Ch 13: Ch5 |  |
| Ch 14: Ch6 |  |
| Ch 15: Ch7 |  |
| Ch 16: Ch8 |  |
| Ch 17: Ch9 |  |

## ANSWER KEY

# Chapter 1: Ch10

Time Allowed: 2 Hours | Total Marks: 100

## SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The mathematical instruments used to construct perpendiculars in class are
  - A. a protractor and a calculator
  - B. a pair of compasses and a ruler
  - C. a measuring tape and a balance
  - D. a clock and a set square
2. Two lines are perpendicular if the angle between them is
  - A.  $90^\circ$
  - B.  $45^\circ$
  - C.  $180^\circ$
  - D.  $60^\circ$
3. To construct an exact copy of a line segment PQ, we use
  - A. a protractor only
  - B. a balance
  - C. a pair of compasses and a ruler
  - D. a calculator
4. To drop a perpendicular means to
  - A. draw a line parallel to the given line
  - B. erect a perpendicular at an end point of the line
  - C. extend the given line
  - D. draw a perpendicular from a point outside the line to meet the line
5. The symbol used to show that two lines are perpendicular is
  - A.  $\parallel$
  - B.  $\perp$
  - C.  $\angle$
  - D.  $\angle$
6. A perpendicular PS is erected at a point P on the line segment AB. The angle APS measures
  - A.  $45^\circ$
  - B.  $180^\circ$
  - C.  $60^\circ$
  - D.  $90^\circ$
7. To copy a line segment PQ onto a ray starting from a point R, we
  - A. open the compasses to the length PQ and draw an arc from R to cut the ray
  - B. use a protractor to mark the length
  - C. estimate the middle of PQ
  - D. draw a line parallel to PQ
8. A line perpendicular to a horizontal line is
  - A. also horizontal
  - B. at  $45^\circ$  to it
  - C. vertical
  - D. curved
9. The perpendicular bisector of a line segment cuts the segment at
  - A. one end
  - B. its midpoint
  - C. any point
  - D. one quarter of its length
10. A line segment  $AB = 8$  cm is bisected perpendicularly. Each half measures
  - A. 16 cm
  - B. 2 cm
  - C. 8 cm
  - D. 4 cm
11. When constructing the perpendicular bisector of AB, the compasses should be opened to a width
  - A. more than half of AB
  - B. exactly half of AB

- C. less than half of AB
- D. exactly equal to AB

- 12.** A line segment  $CD = 5.5$  cm is bisected perpendicularly. Each half measures
- A. 2.25 cm
  - B. 2.75 cm

- C. 5.5 cm  
D. 11 cm
- 13.** In constructing the perpendicular bisector of a segment, the arcs are drawn
- A. from one end point only
  - B. from the midpoint
  - C. from a point outside the line
  - D. from both end points with equal radii
- 15.** A line segment is 12.4 cm long. The distance from one end to the point where its perpendicular bisector meets it is
- A. 12.4 cm
  - B. 24.8 cm
  - C. 6.2 cm
  - D. 3.1 cm
- 17.** An angle bisector divides an angle into
- A. two unequal parts
  - B. three equal parts
  - C. two parts of  $90^\circ$  each
  - D. two equal parts
- 19.** To construct  $60^\circ$  with a pair of compasses, we essentially construct
- A. a square
  - B. a right angle and halve it
  - C. an equilateral triangle
  - D. a perpendicular bisector
- 21.** Bisecting an angle of  $30^\circ$  gives
- A.  $20^\circ$
  - B.  $10^\circ$
  - C.  $15^\circ$
  - D.  $45^\circ$
- 23.** An angle of  $45^\circ$  can be constructed by
- A. bisecting  $60^\circ$
  - B. bisecting  $90^\circ$
  - C. bisecting  $30^\circ$
  - D. copying  $50^\circ$
- 25.** To construct an angle of  $15^\circ$ , we bisect an angle of
- A.  $45^\circ$
  - B.  $60^\circ$
  - C.  $30^\circ$
  - D.  $90^\circ$
- 14.** Any point on the perpendicular bisector of AB is
- A. equidistant from A and B
  - B. always nearer to A than to B
  - C. twice as far from B as from A
  - D. always at the midpoint of AB
- 16.** When constructing a perpendicular bisector, the two arcs cross
- A. on the line segment
  - B. on either side of the line segment
  - C. only above the line
  - D. at the end points
- 18.** Bisecting an angle of  $90^\circ$  gives two angles of
- A.  $45^\circ$
  - B.  $30^\circ$
  - C.  $60^\circ$
  - D.  $22.5^\circ$
- 20.** Bisecting an angle of  $60^\circ$  gives
- A.  $30^\circ$
  - B.  $15^\circ$
  - C.  $45^\circ$
  - D.  $20^\circ$
- 22.** An angle of  $75^\circ$  can be constructed by
- A. bisecting the angle between  $60^\circ$  and  $90^\circ$
  - B. bisecting a right angle
  - C. bisecting  $60^\circ$
  - D. bisecting  $30^\circ$
- 24.** Construct  $\angle PQR = 75^\circ$  and bisect it. Each of the two angles obtained measures
- A.  $75^\circ$
  - B.  $150^\circ$
  - C.  $30^\circ$
  - D.  $37.5^\circ$
- 26.** When copying an angle with compasses, what is transferred from the original angle to the new line?
- A. the protractor reading
  - B. the area of the angle
  - C. only the lengths of the arms
  - D. an arc and its chord

- 27.** Bisecting  $\angle ABC = 120^\circ$  gives two angles of  
A.  $240^\circ$   
B.  $60^\circ$   
C.  $90^\circ$   
D.  $30^\circ$
- 29.** Which of the following angles can ALL be constructed using a pair of compasses and a ruler only?  
A.  $30^\circ$ ,  $45^\circ$ ,  $60^\circ$ ,  $75^\circ$  and  $90^\circ$   
B.  $10^\circ$ ,  $20^\circ$  and  $40^\circ$   
C.  $5^\circ$ ,  $35^\circ$  and  $85^\circ$   
D.  $50^\circ$ ,  $70^\circ$  and  $100^\circ$
- 31.** In the construction of  $45^\circ$ , the line that bisects the right angle is called the  
A. angle bisector  
B. perpendicular bisector  
C. hypotenuse  
D. radius
- 33.** Which of the following best illustrates perpendicular lines in the environment?  
A. the curved rim of a bowl  
B. the corner of a classroom wall  
C. the surface of a ball  
D. the arc of a rainbow
- 35.** Which everyday situation suggests an angle bisector?  
A. the crease formed when a paper corner is folded so that its two edges match  
B. a torn edge of paper  
C. a circular plate  
D. a straight ruler
- 37.** A roof frame is symmetric about its central member, which divides the roof angle into two equal parts. The central member acts as  
A. a perpendicular  
B. a parallel line  
C. a chord  
D. an angle bisector
- 28.** Construct  $\angle XYZ = 60^\circ$  and bisect it, then bisect one of the angles obtained again. The smallest angle obtained is  
A.  $30^\circ$   
B.  $20^\circ$   
C.  $15^\circ$   
D.  $45^\circ$
- 30.** After constructing  $90^\circ$  by raising a perpendicular at a point on a line, the angle is verified using  
A. a ruler  
B. the compasses  
C. a balance  
D. a protractor
- 32.** Construct  $\angle AOB = 60^\circ$  and draw its bisector OC. The angles AOC and COB each measure  
A.  $60^\circ$   
B.  $45^\circ$   
C.  $30^\circ$   
D.  $120^\circ$
- 34.** A carpenter wants the two sides of a window frame to meet at right angles. The construction idea being applied is  
A. angle bisection  
B. perpendicular bisector of a line  
C. copying a line segment  
D. perpendicular lines
- 36.** A rectangular box (crate) has adjacent sides that meet at  
A. acute angles  
B. right angles  
C. obtuse angles  
D. reflex angles
- 38.** Estimating the size of the angle opened by a pair of scissors is an example of  
A. estimating angles in everyday tools  
B. constructing  $60^\circ$   
C. bisecting a line segment  
D. measuring mass

**39.** Which of the following pairs of objects meet at right angles?

- A. two parallel railway lines
- B. the opposite edges of a blackboard
- C. the floor and the wall of a classroom
- D. the two long edges of a ruler

**40.** Constructions such as perpendiculars and bisectors are useful in daily life because they help us to

- A. count money accurately
- B. design and check structures such as buildings and boxes
- C. memorise multiplication tables
- D. weigh objects

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

(a) Perpendiculars and copying line segments.

### **Question 2**

(a) Perpendicular bisectors of line segments.

### **Question 3**

(a) Copying and bisecting angles.

### **Question 4**

(a) Constructing  $90^\circ$ ,  $45^\circ$ ,  $60^\circ$  and  $30^\circ$ .

### **Question 5**

(a) Constructing  $15^\circ$  and  $75^\circ$ .

### **Question 6**

(a) Constructions in the environment.

## Chapter 2: Ch11

Time Allowed: 2 Hours | Total Marks: 100

### SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Write 7 cm 5 mm in centimetres.
  - A. 7.05 cm
  - B. 75 cm
  - C. 7.5 cm
  - D. 12 cm
2. Write 1 m 75 cm in metres.
  - A. 1.075 m
  - B. 1.75 m
  - C. 17.5 m
  - D. 76 m
3. Write 2 km 300 m in metres.
  - A. 500 m
  - B. 230 m
  - C. 23 000 m
  - D. 2300 m
4. The sides of a triangle are 3 cm, 4 cm 5 mm and 5 cm 5 mm. Its perimeter is
  - A. 13 cm
  - B. 12 cm
  - C. 13.5 cm
  - D. 15 cm
5. A rectangle has length 2 m 40 cm and width 1 m 60 cm. Its perimeter is
  - A. 4 m
  - B. 16 m
  - C. 8 m
  - D. 6 m
6. Each side of a square measures 5 cm 4 mm. The perimeter of the square is
  - A. 21.6 cm
  - B. 9.4 cm
  - C. 20.16 cm
  - D. 22 cm
7. Write 4500 m in kilometres.
  - A. 45 km
  - B. 0.45 km
  - C. 450 km
  - D. 4.5 km
8. A rectangular field measures 120 m by 80 m. The distance round the field is
  - A. 200 m
  - B. 400 m
  - C. 9600 m
  - D. 320 m
9. The perimeter of an equilateral triangle of side 4 cm 2 mm is
  - A. 12.2 cm
  - B. 16.8 cm
  - C. 12.6 cm
  - D. 8.6 cm
10. Find the length of wire needed to fence a square garden of side 25 m 50 cm once round.
  - A. 102 m
  - B. 100 m
  - C. 51 m
  - D. 204 m
11. Write 3 m 8 cm in metres.
  - A. 3.8 m
  - B. 30.8 m
  - C. 3.18 m
  - D. 3.08 m
12. The four sides of a field measure 1 km 200 m, 800 m, 1 km and 1 km. The perimeter of the field, in kilometres, is
  - A. 3 km
  - B. 4 km

- C. 4.2 km
- D. 40 km

- 14.** The distance from the centre of a circle to any point on the circle is the
- A. diameter
  - B. circumference
  - C. radius
  - D. chord

- 13.** The distance round a circle is called the
- A. circumference
  - B. radius
  - C. diameter
  - D. arc

- 15.** A straight line passing through the centre of a circle and joining two points on the circle is the
- A. radius
  - B. tangent
  - C. arc
  - D. diameter
- 17.** The ratio of the circumference of a circle to its diameter is always
- A.  $\pi$  (pi)
  - B. 2
  - C. 4
  - D. the radius
- 19.** The circumference  $C$  of a circle of radius  $r$  is given by
- A.  $C = \pi r^2$
  - B.  $C = \pi r$
  - C.  $C = 2\pi + r$
  - D.  $C = 2\pi r$
- 21.** Find the circumference of a circle of diameter 10 cm, rounding your answer to the nearest tenth. [Take  $\pi = 3.142$ ]
- A. 31.4 cm
  - B. 62.8 cm
  - C. 314.2 cm
  - D. 3.14 cm
- 23.** The diameter of a circle is 140 cm. Its radius is
- A. 280 cm
  - B. 70 cm
  - C. 140 cm
  - D. 35 cm
- 25.** The radius of a circle is 140 cm. Its diameter is
- A. 280 cm
  - B. 70 cm
  - C. 440 cm
  - D. 140 cm
- 27.** Which of the following can be used to measure the circumference of a can directly?
- A. a balance
  - B. a protractor
  - C. a straight ruler alone
  - D. a piece of string and a ruler
- 16.** If  $r$  is the radius and  $d$  the diameter of a circle, then
- A.  $d = r$
  - B.  $d = 2r$
  - C.  $d = r \div 2$
  - D.  $d = r + 2$
- 18.** A good fractional approximation of  $\pi$  used in schools is
- A.  $\frac{22}{5}$
  - B.  $\frac{21}{7}$
  - C.  $\frac{22}{7}$
  - D.  $\frac{7}{22}$
- 20.** Find the circumference of a circle of radius 7 cm. [Take  $\pi = \frac{22}{7}$ ]
- A. 22 cm
  - B. 44 cm
  - C. 154 cm
  - D. 88 cm
- 22.** The circumference of a circle is 88 cm. Find its diameter. [Take  $\pi = \frac{22}{7}$ ]
- A. 14 cm
  - B. 44 cm
  - C. 28 cm
  - D. 176 cm
- 24.** The radius of a circle is 140 cm. Find its circumference. [Take  $\pi = \frac{22}{7}$ ]
- A. 440 cm
  - B. 1760 cm
  - C. 220 cm
  - D. 880 cm
- 26.** A tin lid has circumference 38 cm and diameter 12 cm. The value of  $c \div d$ , correct to 2 decimal places, is
- A. 0.32
  - B. 26
  - C. 3.17
  - D. 3.8

- 29.** The region of a circle enclosed by two radii and an arc is called
- A. a chord
  - B. a sector
  - C. a segment
  - D. a tangent
- 31.** A bicycle wheel has radius 35 cm. The distance covered in one complete turn of the wheel is [Take  $\pi = 22/7$ ]
- A. 110 cm
  - B. 70 cm
  - C. 440 cm
  - D. 220 cm
- 33.** On a square grid, the area of a rectangle is found by
- A. length + width
  - B.  $2 \times (\text{length} + \text{width})$
  - C.  $4 \times \text{length}$
  - D. length  $\times$  width
- 35.** A rectangle drawn on a square grid is 4 units by 3 units. Another rectangle is drawn with twice the area. The area of the new rectangle is
- A. 6 square units
  - B. 12 square units
  - C. 24 square units
  - D. 48 square units
- 37.** The area of a triangle is given by
- A.  $\frac{1}{2} \times \text{base} \times \text{height}$
  - B. base  $\times$  height
  - C.  $2 \times \text{base} \times \text{height}$
  - D. base + height
- 39.** Which base-and-height pair can be used to draw a triangle of area 3 square units on a square grid?
- A. base 3 units, height 3 units
  - B. base 2 units, height 2 units
  - C. base 6 units, height 2 units
  - D. base 6 units, height 1 unit
- 28.** A part of the circumference of a circle between two points on it is called
- A. an arc
  - B. a sector
  - C. a diameter
  - D. a centre
- 30.** A circular roundabout has radius 14 m. A man walks once round it. The distance he walks is [Take  $\pi = 22/7$ ]
- A. 44 m
  - B. 28 m
  - C. 88 m
  - D. 616 m
- 32.** The circumference of a car wheel is 132 cm. Find the radius of the wheel. [Take  $\pi = 22/7$ ]
- A. 42 cm
  - B. 21 cm
  - C. 84 cm
  - D. 66 cm
- 34.** A rectangle drawn on a square grid is 4 units long and 3 units wide. Its area is
- A. 12 square units
  - B. 7 square units
  - C. 14 square units
  - D. 24 square units
- 36.** A rectangle on a square grid is 4 units wide and 6 units long. A new rectangle is twice as wide and one and a half times as long. Its dimensions are
- A. 6 units wide and 9 units long
  - B. 8 units wide and 9 units long
  - C. 8 units wide and 6 units long
  - D. 4 units wide and 9 units long
- 38.** A triangle has base 6 units and height 4 units. Its area is
- A. 24 square units
  - B. 10 square units
  - C. 12 square units
  - D. 6 square units
- 40.** How many different triangles of the same area can be drawn on a square grid when the base and height stay the same?
- A. many different triangles

- B. only one triangle
- C. exactly two triangles
- D. no triangle

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

(a) Perimeters in mixed units.

### **Question 2**

(a) Parts of a circle and the value of  $\pi$ .

### **Question 3**

(a) Circumference of a circle.

### **Question 4**

(a) Solving problems with circumference.

### **Question 5**

(a) Drawing on square grids.

### **Question 6**

(a) Mixed problems.

## Chapter 3: Ch12

Time Allowed: 2 Hours | Total Marks: 100

### SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. A rectangle is cut along a diagonal to form two triangles. The area of each triangle is
  - A. half the area of the rectangle
  - B. equal to the area of the rectangle
  - C. twice the area of the rectangle
  - D. one quarter of the area of the rectangle
3. A triangle has base 8 cm and perpendicular height 5 cm. The area of the rectangle that encloses it is
  - A.  $20 \text{ cm}^2$
  - B.  $40 \text{ cm}^2$
  - C.  $13 \text{ cm}^2$
  - D.  $80 \text{ cm}^2$
5. In the formula  $\frac{1}{2} \times \text{base} \times \text{height}$ , the height used is the
  - A. perpendicular height
  - B. length of a slant side
  - C. perimeter
  - D. longest side
7. The formula for the area of a triangle includes  $\frac{1}{2}$  because
  - A. triangles are always small
  - B. a triangle is half of a rectangle or parallelogram with the same base and height
  - C. areas are always halved
  - D. the height is measured twice
9. Two identical triangles can be fitted together to form
  - A. a parallelogram
  - B. a circle
  - C. a pentagon
  - D. a cube
11. Which of the following is NOT needed to find the area of a triangle?
  - A. its base
2. The area of a triangle is given by
  - A.  $\text{base} \times \text{height}$
  - B.  $2 \times \text{base} \times \text{height}$
  - C.  $\text{base} + \text{height}$
  - D.  $\frac{1}{2} \times \text{base} \times \text{height}$
4. The area of the triangle in Question 3 is
  - A.  $40 \text{ cm}^2$
  - B.  $13 \text{ cm}^2$
  - C.  $20 \text{ cm}^2$
  - D.  $80 \text{ cm}^2$
6. A triangle and a parallelogram have the same base and the same height. The area of the triangle is
  - A. equal to that of the parallelogram
  - B. twice that of the parallelogram
  - C. half that of the parallelogram
  - D. one quarter of that of the parallelogram
8. The area of a parallelogram with base  $b$  and height  $h$  is
  - A.  $\frac{1}{2} b h$
  - B.  $b + h$
  - C.  $2 b h$
  - D.  $b h$
10. A triangle drawn on square grid encloses 6 whole unit squares and 8 half unit squares. Its area is
  - A. 14 square units
  - B. 12 square units
  - C. 10 square units
  - D. 8 square units

- B. its perpendicular height
- C. the factor  $\frac{1}{2}$
- D. the length of its slant side

- 12.** The area of a right-angled triangle whose two perpendicular sides are 6 cm and 8 cm is
- A.  $24 \text{ cm}^2$
  - B.  $48 \text{ cm}^2$
  - C.  $14 \text{ cm}^2$

- 13.** Find the area of a triangle with base 10 cm and height 6 cm.
- A.  $60 \text{ cm}^2$
  - B.  $16 \text{ cm}^2$
  - C.  $30 \text{ cm}^2$
  - D.  $32 \text{ cm}^2$
- 15.** Find the area of a triangle with base 7 cm and height 4 cm.
- A.  $28 \text{ cm}^2$
  - B.  $11 \text{ cm}^2$
  - C.  $22 \text{ cm}^2$
  - D.  $14 \text{ cm}^2$
- 17.** The area of a triangle is  $24 \text{ cm}^2$  and its base is 8 cm. Its perpendicular height is
- A. 3 cm
  - B. 12 cm
  - C. 6 cm
  - D. 4 cm
- 19.** Find the area of a triangle with base 4.2 cm and height 5 cm.
- A.  $21 \text{ cm}^2$
  - B.  $9.2 \text{ cm}^2$
  - C.  $42 \text{ cm}^2$
  - D.  $10.5 \text{ cm}^2$
- 21.** A triangle has area  $36 \text{ cm}^2$  and base 9 cm. Its perpendicular height is
- A. 8 cm
  - B. 4 cm
  - C. 16 cm
  - D. 6 cm
- 23.** The area of a right-angled triangle whose two perpendicular sides are 5 cm and 12 cm is
- A.  $60 \text{ cm}^2$
  - B.  $30 \text{ cm}^2$
  - C.  $17 \text{ cm}^2$
  - D.  $34 \text{ cm}^2$
- 25.** A triangle has area  $42 \text{ cm}^2$  and perpendicular height 7 cm. Its base is
- A. 6 cm
  - B. 21 cm
  - C. 12 cm
  - D.  $28 \text{ cm}^2$
- 14.** Find the area of a triangle with base 12 cm and height 9 cm.
- A.  $108 \text{ cm}^2$
  - B.  $54 \text{ cm}^2$
  - C.  $21 \text{ cm}^2$
  - D.  $42 \text{ cm}^2$
- 16.** A triangular garden has base 15 m and height 8 m. Its area is
- A.  $120 \text{ m}^2$
  - B.  $60 \text{ m}^2$
  - C.  $23 \text{ m}^2$
  - D.  $46 \text{ m}^2$
- 18.** The area of a triangle is  $50 \text{ cm}^2$  and its perpendicular height is 10 cm. Its base is
- A. 10 cm
  - B. 5 cm
  - C. 25 cm
  - D. 100 cm
- 20.** Find the area of a triangle with base 9 cm and height 6 cm.
- A.  $54 \text{ cm}^2$
  - B.  $27 \text{ cm}^2$
  - C.  $15 \text{ cm}^2$
  - D.  $30 \text{ cm}^2$
- 22.** Find the area of a triangle with base 3.5 cm and height 4 cm.
- A.  $14 \text{ cm}^2$
  - B.  $7.5 \text{ cm}^2$
  - C.  $3.5 \text{ cm}^2$
  - D.  $7 \text{ cm}^2$
- 24.** A triangular flower bed has base 20 m and height 15 m. Its area is
- A.  $150 \text{ m}^2$
  - B.  $300 \text{ m}^2$
  - C.  $35 \text{ m}^2$
  - D.  $70 \text{ m}^2$

D. 24 cm

**27.** A triangular flag has base 30 cm and height 20 cm. The area of cloth needed for the flag is

- A.  $600 \text{ cm}^2$
- B.  $300 \text{ cm}^2$
- C.  $50 \text{ cm}^2$
- D.  $100 \text{ cm}^2$

**26.** A triangle has area  $10 \text{ cm}^2$ . If both its base and its height are doubled, the new area is

- A.  $20 \text{ cm}^2$
- B.  $10 \text{ cm}^2$
- C.  $15 \text{ cm}^2$
- D.  $40 \text{ cm}^2$

- 29.** A triangular piece of metal has area  $60 \text{ cm}^2$  and base  $12 \text{ cm}$ . Its perpendicular height is
- A.  $5 \text{ cm}$
  - B.  $24 \text{ cm}$
  - C.  $6 \text{ cm}$
  - D.  $10 \text{ cm}$
- 31.** A square of side  $8 \text{ cm}$  is cut along a diagonal. The area of each triangle formed is
- A.  $64 \text{ cm}^2$
  - B.  $32 \text{ cm}^2$
  - C.  $16 \text{ cm}^2$
  - D.  $8 \text{ cm}^2$
- 33.** Two triangles each have base  $6 \text{ cm}$  and height  $4 \text{ cm}$ . Their total area is
- A.  $12 \text{ cm}^2$
  - B.  $48 \text{ cm}^2$
  - C.  $20 \text{ cm}^2$
  - D.  $24 \text{ cm}^2$
- 35.** A shape is made of a rectangle  $10 \text{ cm}$  by  $4 \text{ cm}$  with a triangle of base  $10 \text{ cm}$  and height  $3 \text{ cm}$  on top. The total area is
- A.  $40 \text{ cm}^2$
  - B.  $55 \text{ cm}^2$
  - C.  $70 \text{ cm}^2$
  - D.  $47.5 \text{ cm}^2$
- 37.** A farmer has a triangular field with base  $60 \text{ m}$  and height  $50 \text{ m}$ . The area of the field is
- A.  $3000 \text{ m}^2$
  - B.  $110 \text{ m}^2$
  - C.  $1500 \text{ m}^2$
  - D.  $220 \text{ m}^2$
- 39.** Two triangles have the same base and the same height. They must
- A. be congruent
  - B. have equal areas
  - C. have equal perimeters
  - D. be right-angled
- 28.** A triangular plot of land has base  $40 \text{ m}$  and height  $25 \text{ m}$ . Its area is
- A.  $1000 \text{ m}^2$
  - B.  $65 \text{ m}^2$
  - C.  $500 \text{ m}^2$
  - D.  $130 \text{ m}^2$
- 30.** A triangular wall has base  $10 \text{ m}$  and height  $6 \text{ m}$ . Paint costs GH¢2.00 per square metre. The cost of painting the wall is
- A. GH¢60.00
  - B. GH¢30.00
  - C. GH¢120.00
  - D. GH¢16.00
- 32.** A rectangle  $12 \text{ cm}$  by  $5 \text{ cm}$  is cut along a diagonal. The area of each triangle is
- A.  $30 \text{ cm}^2$
  - B.  $60 \text{ cm}^2$
  - C.  $17 \text{ cm}^2$
  - D.  $25 \text{ cm}^2$
- 34.** The triangular sail of a boat has base  $4 \text{ m}$  and height  $3 \text{ m}$ . The area of the sail is
- A.  $12 \text{ m}^2$
  - B.  $7 \text{ m}^2$
  - C.  $6 \text{ m}^2$
  - D.  $24 \text{ m}^2$
- 36.** A triangle has area  $27 \text{ cm}^2$  and perpendicular height  $6 \text{ cm}$ . Its base is
- A.  $9 \text{ cm}$
  - B.  $4.5 \text{ cm}$
  - C.  $18 \text{ cm}$
  - D.  $3 \text{ cm}$
- 38.** Four identical triangular flags, each of base  $20 \text{ cm}$  and height  $15 \text{ cm}$ , are to be made. The total area of cloth needed is
- A.  $150 \text{ cm}^2$
  - B.  $300 \text{ cm}^2$
  - C.  $1200 \text{ cm}^2$
  - D.  $600 \text{ cm}^2$
- 40.** The area of a triangle with base  $2.4 \text{ m}$  and height  $5 \text{ m}$  is
- A.  $6 \text{ m}^2$
  - B.  $12 \text{ m}^2$
  - C.  $7.4 \text{ m}^2$
  - D.  $3 \text{ m}^2$

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

(a) Deriving the formula for the area of a triangle.

### **Question 2**

(a) Calculating areas of triangles.

### **Question 3**

(a) Finding missing dimensions.

### **Question 4**

(a) Areas on square grids.

### **Question 5**

(a) Area problems in real life.

### **Question 6**

(a) Creating and reasoning.

## Chapter 4: Ch13

Time Allowed: 2 Hours | Total Marks: 100

### SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Bearings are always measured from the direction
  - A. East
  - B. South
  - C. West
  - D. North
2. Bearings are measured in the
  - A. clockwise direction
  - B. anti-clockwise direction
  - C. either direction
  - D. vertical direction
3. A bearing is written as a
  - A. one-digit number
  - B. two-digit number
  - C. three-digit number
  - D. four-digit number
4. The bearing of a point due East of another point is
  - A.  $045^\circ$
  - B.  $090^\circ$
  - C.  $180^\circ$
  - D.  $270^\circ$
5. The bearing of a point due South of another point is
  - A.  $090^\circ$
  - B.  $270^\circ$
  - C.  $300^\circ$
  - D.  $180^\circ$
6. The bearing of a point due West of another point is
  - A.  $270^\circ$
  - B.  $180^\circ$
  - C.  $090^\circ$
  - D.  $360^\circ$
7. The bearing of a point due North of another point is written as
  - A.  $090^\circ$
  - B.  $000^\circ$
  - C.  $180^\circ$
  - D.  $045^\circ$
8. From a point O, a point P is  $60^\circ$  measured clockwise from the North. The three-figure bearing of P from O is
  - A.  $030^\circ$
  - B.  $300^\circ$
  - C.  $120^\circ$
  - D.  $060^\circ$
9. The bearing of P from Q is  $060^\circ$ . The bearing of Q from P is
  - A.  $120^\circ$
  - B.  $060^\circ$
  - C.  $240^\circ$
  - D.  $300^\circ$
10. The bearing of P from Q is  $145^\circ$ . The bearing of Q from P is
  - A.  $325^\circ$
  - B.  $035^\circ$
  - C.  $215^\circ$
  - D.  $055^\circ$
11. The bearing of A from B is  $200^\circ$ . The bearing of B from A is
  - A.  $380^\circ$
  - B.  $100^\circ$
  - C.  $020^\circ$
  - D.  $110^\circ$
12. The back bearing of a bearing less than  $180^\circ$  is found by
  - A. subtracting it from  $360^\circ$
  - B. multiplying it by 2
  - C. subtracting  $90^\circ$
  - D. adding  $180^\circ$

- 13.** The back bearing of a bearing greater than  $180^\circ$  is found by
- subtracting  $180^\circ$
  - adding  $180^\circ$
  - subtracting it from  $90^\circ$
  - adding  $90^\circ$
- 14.** The bearing of X from Y is  $275^\circ$ . The bearing of Y from X is
- $455^\circ$
  - $095^\circ$
  - $085^\circ$
  - $105^\circ$
- 15.** The bearing of B from A is  $040^\circ$  and  $AB = 5$  km. This information describes
- a scalar quantity only
  - the speed of travel
  - the position of B relative to A
  - the area between A and B
- 16.** Two towns P and Q are 8 km apart. The bearing of Q from P is  $120^\circ$ . The bearing of P from Q is
- $060^\circ$
  - $300^\circ$
  - $240^\circ$
  - $120^\circ$
- 17.** A vector quantity has
- magnitude and direction
  - magnitude only
  - direction only
  - neither magnitude nor direction
- 18.** Which of the following is a vector quantity?
- speed
  - distance
  - velocity
  - mass
- 19.** Which of the following is a scalar quantity?
- force
  - weight
  - velocity
  - time
- 20.** Which list contains scalar quantities only?
- force, mass, time
  - distance, speed, mass
  - velocity, energy, work
  - weight, momentum, volume
- 21.** Which list contains vector quantities only?
- weight, force, velocity, momentum
  - time, force, speed, mass
  - distance, energy, velocity, work
  - mass, volume, weight, energy
- 22.** A vector may be identified as a movement (distance) along
- any curved path
  - the North line only
  - a given bearing
  - a circle
- 23.** The vector (3 km,  $060^\circ$ ) means
- 3 km at a bearing of  $030^\circ$
  - a movement of 3 km on a bearing of  $060^\circ$
  - 60 km on a bearing of  $003^\circ$
  - 3 km due North
- 24.** In the vector (5 km,  $120^\circ$ ), the direction of movement is on a bearing of
- $060^\circ$
  - $030^\circ$
  - $210^\circ$
  - $120^\circ$
- 25.** The column form of a vector with x-component 2 and y-component 3 is
- (2, 3)
  - (3, 2)
  - (5, 0)
  - (2, -3)
- 26.** In the column vector (x, y), the x-component represents movement in the
- vertical direction
  - direction of the bearing
  - magnitude of the vector
  - horizontal (East-West) direction

- 27.** In the column vector  $(x, y)$ , the y-component represents movement in the
- A. horizontal direction
  - B. vertical (North-South) direction
  - C. speed of travel
  - D. total distance
- 29.** The magnitude of the vector  $(3, 4)$  is
- A. 7
  - B. 12
  - C. 25
  - D. 5
- 31.** The magnitude of the vector  $(5, 12)$  is
- A. 17
  - B. 13
  - C. 60
  - D. 169
- 33.** The magnitude of a column vector is found using
- A. a protractor
  - B. the compass rose
  - C. simple addition
  - D. Pythagoras' theorem
- 35.** The magnitude of the vector  $(8, 15)$  is
- A. 23
  - B. 17
  - C. 120
  - D. 7
- 37.** A boat sails 6 km North and then 8 km East. Its straight-line distance from the starting point is
- A. 14 km
  - B. 48 km
  - C. 2 km
  - D. 10 km
- 39.** A girl walks 4 km North and then 3 km East. Her straight-line distance from the start is
- A. 7 km
  - B. 12 km
  - C. 5 km
  - D. 1 km
- 28.** The column vector  $(-3, 4)$  represents a movement of
- A. 3 units East and 4 units North
  - B. 3 units West and 4 units South
  - C. 3 units West and 4 units North
  - D. 4 units West and 3 units North
- 30.** The magnitude of the vector  $(6, 8)$  is
- A. 10
  - B. 14
  - C. 48
  - D. 100
- 32.** The magnitude of the vector  $(-3, 4)$  is
- A. 1
  - B. -5
  - C. 5
  - D. 7
- 34.** The magnitude of the vector  $(9, 12)$  is
- A. 15
  - B. 21
  - C. 108
  - D. 225
- 36.** The magnitude of the vector  $(2, 3)$ , correct to 1 decimal place, is
- A. 3.6
  - B. 5
  - C. 13
  - D. 2.5
- 38.** A man walks 9 km East and then 12 km South. His straight-line distance from the start is
- A. 21 km
  - B. 15 km
  - C. 108 km
  - D. 3 km
- 40.** The magnitude of the vector  $(7, 24)$  is
- A. 25
  - B. 31
  - C. 168
  - D. 625

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

(a) Bearings.

### **Question 2**

(a) Back bearings.

### **Question 3**

(a) Scalar and vector quantities.

### **Question 4**

(a) Vectors in column form.

### **Question 5**

(a) Magnitude of vectors.

### **Question 6**

(a) Mixed problems.

## Chapter 5: Ch14

Time Allowed: 2 Hours | Total Marks: 100

### SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. A shape has reflectional (fold) symmetry if it can be
  - A. moved to a new position
  - B. folded into two matching halves
  - C. turned through any angle
  - D. made bigger
2. The number of lines of symmetry of a square is
  - A. 1
  - B. 2
  - C. 4
  - D. 8
3. The number of lines of symmetry of a rectangle (which is not a square) is
  - A. 2
  - B. 4
  - C. 1
  - D. 0
4. Which of the following has NO line of symmetry?
  - A. an equilateral triangle
  - B. an isosceles triangle
  - C. a circle
  - D. a scalene triangle
5. Many adinkra symbols are examples of designs with
  - A. translation only
  - B. reflectional symmetry
  - C. no symmetry at all
  - D. curved edges only
6. The number of lines of symmetry of a circle is
  - A. 1
  - B. 4
  - C. 0
  - D. unlimited
7. The number of lines of symmetry of an equilateral triangle is
  - A. 3
  - B. 0
  - C. 1
  - D. 2
8. The fold line that divides a symmetric shape into two matching halves is called a
  - A. diagonal
  - B. line of symmetry
  - C. radius
  - D. tangent
9. The reflection of the point (3, 1) in the x-axis is
  - A. (-3, 1)
  - B. (1, 3)
  - C. (3, -1)
  - D. (-3, -1)
10. The reflection of the point (2, 5) in the y-axis is
  - A. (2, -5)
  - B. (5, 2)
  - C. (-2, -5)
  - D. (-2, 5)
11. Under reflection, a shape and its image are always
  - A. congruent
  - B. of different sizes
  - C. of different shapes
  - D. rotated
12. The reflection of the point (4, 3) in the x-axis is
  - A. (3, 4)
  - B. (-4, 3)
  - C. (4, -3)
  - D. (-4, -3)

- 13.** The reflection of the point  $(-2, 3)$  in the  $y$ -axis is
- $(-2, -3)$
  - $(3, -2)$
  - $(-2, 3)$
  - $(2, 3)$
- 15.** A point and its reflection in a mirror line are
- equal distances from the mirror line on opposite sides
  - on the same side of the mirror line
  - at different distances from the mirror line
  - both on the mirror line
- 17.** The point  $A(3, 1)$  is reflected in the  $x$ -axis. The image of  $A$  is
- $(1, 3)$
  - $(3, -1)$
  - $(-3, 1)$
  - $(-3, -1)$
- 19.** The reflection of the point  $(0, 5)$  in the  $x$ -axis is
- $(5, 0)$
  - $(0, 5)$
  - $(5, 5)$
  - $(0, -5)$
- 21.** A translation is
- flipping a shape over a line
  - turning a shape about a point
  - changing the size of a shape
  - sliding a shape without turning or flipping it
- 23.** The point  $(2, 3)$  is translated by the vector  $(4, 1)$ . Its image is
- $(6, 3)$
  - $(8, 4)$
  - $(6, 4)$
  - $(-2, 2)$
- 25.** The point  $(1, 1)$  is translated by the vector  $(2, 5)$ . Its image is
- $(3, 5)$
  - $(2, 5)$
  - $(5, 2)$
  - $(3, 6)$
- 14.** A point that lies on the mirror line, when reflected,
- moves to the opposite side
  - stays in the same position
  - moves twice as far
  - becomes the origin
- 16.** A point is 3 units to the right of the origin and 2 units up. Its coordinates are
- $(2, 3)$
  - $(-3, 2)$
  - $(3, 2)$
  - $(3, -2)$
- 18.** The triangle with vertices  $(1, 1)$ ,  $(3, 1)$  and  $(1, 4)$  is reflected in the  $y$ -axis. The vertices of the image are
- $(-1, 1)$ ,  $(-3, 1)$ ,  $(-1, 4)$
  - $(1, -1)$ ,  $(3, -1)$ ,  $(1, -4)$
  - $(-1, -1)$ ,  $(-3, -1)$ ,  $(-1, -4)$
  - $(1, 1)$ ,  $(3, 1)$ ,  $(1, 4)$
- 20.** Reflection in the  $x$ -axis changes
- both coordinates
  - neither coordinate
  - the sign of the  $y$ -coordinate
  - the sign of the  $x$ -coordinate
- 22.** People going down a slide is a real-life example of
- translation
  - reflection
  - enlargement
  - symmetry
- 24.** The point  $(5, 2)$  is translated by the vector  $(-3, 2)$ . Its image is
- $(8, 4)$
  - $(2, 4)$
  - $(2, 0)$
  - $(-3, 2)$
- 26.** The point  $(4, -2)$  is translated by the vector  $(-1, 3)$ . Its image is
- $(3, 1)$
  - $(5, 1)$
  - $(3, -5)$
  - $(-1, 3)$

- 27.** A shape translated by the vector  $(0, 4)$  moves
- 4 units left
  - 4 units up
  - 4 units right
  - 4 units down
- 29.** The translation that moves the point  $(2, 2)$  to  $(5, 6)$  is the vector
- $(3, 4)$
  - $(7, 8)$
  - $(-3, -4)$
  - $(4, 3)$
- 31.** Congruent shapes are shapes which have
- the same shape only
  - the same size only
  - different shapes
  - the same shape and the same size
- 33.** Which of the following statements is true?
- All congruent shapes are similar
  - All similar shapes are congruent
  - Similar shapes must be congruent
  - Congruent shapes have different sizes
- 35.** A rectangle 4 cm by 2 cm and a rectangle 8 cm by 4 cm are
- congruent
  - neither similar nor congruent
  - congruent but not similar
  - similar but not congruent
- 37.** A photograph is enlarged by a photocopier. The enlargement and the original photograph are
- congruent
  - neither similar nor congruent
  - different in shape
  - similar
- 39.** A triangle with sides 2 cm, 3 cm and 4 cm is similar to a triangle with sides
- 2 cm, 3 cm, 5 cm
  - 3 cm, 4 cm, 5 cm
  - 4 cm, 6 cm, 8 cm
  - 2 cm, 4 cm, 6 cm
- 28.** After a translation, the image is
- a different shape from the object
  - smaller than the object
  - bigger than the object
  - congruent to the object
- 30.** The translation that moves the point  $(3, 5)$  to  $(1, 2)$  is the vector
- $(2, 3)$
  - $(4, 7)$
  - $(-2, -3)$
  - $(-3, -2)$
- 32.** Similar shapes are shapes which have
- the same size
  - different shapes
  - the same shape but different sizes
  - the same shape and the same size
- 34.** Carpet designs and fabric patterns often contain
- only single shapes
  - repeated congruent shapes
  - no shapes at all
  - circles only
- 36.** Two circles of different radii are always
- similar
  - congruent
  - neither similar nor congruent
  - equal in area
- 38.** Two triangles are congruent. One has sides 3 cm, 4 cm and 5 cm. The sides of the other are
- 6 cm, 8 cm, 10 cm
  - 3 cm, 4 cm, 5 cm
  - 4 cm, 5 cm, 6 cm
  - 3 cm, 3 cm, 3 cm
- 40.** Under reflection and under translation, the object and its image are always
- congruent
  - similar but not congruent
  - of different sizes
  - turned

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

(a) Reflectional (fold) symmetry.

### **Question 2**

(a) Reflection in the coordinate plane.

### **Question 3**

(a) Translation by vectors.

### **Question 4**

(a) Congruent and similar shapes.

### **Question 5**

(a) Mixed coordinate transformations.

### **Question 6**

(a) Creating and reasoning.

## Chapter 6: Ch15

Time Allowed: 2 Hours | Total Marks: 100

### SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Data that can be counted or measured and expressed as numbers is called
  - A. qualitative data
  - B. quantitative data
  - C. grouped data
  - D. secondary data
2. The favourite subject of a JHS 1 student is an example of
  - A. quantitative data
  - B. continuous data
  - C. qualitative data
  - D. discrete data
3. A class wants to find the favourite drink of its students so it can buy drinks for a party. The best method of collecting this data is a
  - A. observation
  - B. experiment
  - C. database search
  - D. questionnaire
4. The best method of finding the number of desks in each classroom of a school is
  - A. observation and counting
  - B. interview
  - C. questionnaire
  - D. internet search
5. Collecting data by watching people or events and recording what happens, without asking questions, is called
  - A. experiment
  - B. observation
  - C. database search
  - D. survey
6. Which method of data collection makes use of already recorded information such as registers, books or the internet?
  - A. database / electronic media
  - B. observation
  - C. interview
  - D. experiment
7. Which of the following is quantitative data?
  - A. the favourite hobby of a student
  - B. the favourite day of the week
  - C. opinions about school meals
  - D. the height of students in a class
8. To find out whether people who eat more fufu develop pot belly, the best method of collecting data is a
  - A. survey
  - B. questionnaire
  - C. experiment
  - D. database search
9. Data collected directly from the people concerned by asking them questions is called
  - A. secondary data
  - B. grouped data
  - C. qualitative data
  - D. primary data
10. Data obtained from sources such as textbooks, newspapers or the internet, rather than collected firsthand, is called
  - A. primary data
  - B. secondary data
  - C. raw data
  - D. discrete data

- 11.** The best method of finding the amount of money Basic 6 pupils spend on bus fare to school every month is a
- A. questionnaire
  - B. observation
  - C. experiment
  - D. simulation
- 13.** A questionnaire is
- A. a type of graph
  - B. a set of written questions used to collect data from people
  - C. a table of frequencies
  - D. a picture used to represent data
- 15.** A good questionnaire question should be
- A. clear and easy to understand
  - B. leading, pushing the person to one answer
  - C. long and confusing
  - D. two questions joined together
- 17.** Which of the following is the best way to ask about a pupil's favourite snack in a questionnaire?
- A. Write an essay about food
  - B. Describe every meal you have ever eaten
  - C. Tick one: mango, biscuit, groundnut, other
  - D. What about food?
- 19.** After the responses to a questionnaire have been collected, the next step is to
- A. organise and record the results in a table
  - B. throw away the responses
  - C. design another questionnaire
  - D. draw the pie chart first
- 21.** The question "How much do you spend on bus fare to school every day?" collects
- A. qualitative data
  - B. quantitative data
  - C. secondary data
  - D. opinions
- 12.** Which of the following questions will collect quantitative data?
- A. What is your favourite colour?
  - B. What is your favourite subject?
  - C. How many brothers and sisters do you have?
  - D. Do you enjoy homework?
- 14.** Which of the following is a good question for a questionnaire about favourite subjects?
- A. Tell us your whole school history
  - B. Don't you like mathematics?
  - C. What is your favourite subject?
  - D. Is school food nice?
- 16.** The question "Don't you think mathematics is the best subject?" is not a good questionnaire question because it
- A. is too short
  - B. pushes the respondent towards one answer
  - C. cannot be answered at all
  - D. collects only numbers
- 18.** To administer a questionnaire means to
- A. draw a graph of the results
  - B. give the questionnaire to respondents to answer
  - C. count the tally marks
  - D. calculate the sector angles
- 20.** In a class survey, the question "What is your favourite day of the week?" collects
- A. quantitative data
  - B. continuous data
  - C. qualitative data
  - D. grouped data
- 22.** The best way to record the results collected with a questionnaire is in a
- A. frequency (tally) table
  - B. long essay
  - C. list of names only
  - D. photograph

- 23.** Which of the following is NOT a feature of a good questionnaire?
- A. clear questions
  - B. space for the respondent's answers
  - C. leading or ambiguous questions
  - D. short and focused questions
- 25.** When counting with tally marks, the fifth mark is drawn
- A. vertically, next to the others
  - B. horizontally below the group
  - C. diagonally across the four marks
  - D. in a new group far away
- 27.** The total number of values in the data set 4, 2, 4, 3, 2, 4, 1, 3, 4, 2 is
- A. 10
  - B. 8
  - C. 12
  - D. 9
- 29.** Using the frequency table in question 28, the mark obtained by the greatest number of students is
- A. 3
  - B. 5
  - C. 2
  - D. 4
- 31.** The best way to display the change in temperature over a week is a
- A. pie chart
  - B. pictograph
  - C. line graph
  - D. bar chart
- 33.** Which display uses sectors of a circle to represent parts of the data?
- A. bar chart
  - B. line graph
  - C. pictograph
  - D. pie chart
- 35.** In a pictograph, one symbol represents 5 students. How many symbols are needed to represent 35 students?
- A. 5
  - B. 7
  - C. 6
  - D. 8
- 24.** In a survey of 30 students, 12 said Mathematics is their favourite subject. What percentage of the students chose Mathematics?
- A. 12%
  - B. 30%
  - C. 25%
  - D. 40%
- 26.** The following data were recorded: 4, 2, 4, 3, 2, 4, 1, 3, 4, 2. The frequency of the value 4 is
- A. 2
  - B. 3
  - C. 5
  - D. 4
- 28.** In a frequency table, the frequencies of the marks 1, 2, 3, 4 and 5 are 2, 6, 4, 5 and 3 respectively. How many students took the test?
- A. 15
  - B. 20
  - C. 25
  - D. 18
- 30.** Using the frequency table in question 28, the number of students who scored 3 or more is
- A. 12
  - B. 10
  - C. 14
  - D. 15
- 32.** The best way to compare the quantities of different categories of data is a
- A. line graph
  - B. bar chart
  - C. pie chart
  - D. tally table
- 34.** Which display uses pictures or symbols to represent numbers?
- A. pictograph
  - B. pie chart
  - C. bar chart
  - D. line graph
- 36.** In a survey of 40 students, 16 chose football as their favourite sport. The sector angle for football in a pie chart is
- A.  $120^\circ$

- B.  $160^\circ$
- C.  $144^\circ$
- D.  $90^\circ$

**37.** Using the survey in question 36, 10

students chose volleyball. The sector angle for volleyball is

- A.  $100^\circ$
- B.  $72^\circ$
- C.  $54^\circ$
- D.  $90^\circ$

**39.** A double bar chart is used for

- A. counting frequencies
- B. showing sector angles
- C. drawing symbols
- D. comparing two sets of data

**38.** In a bar chart, the height of each bar represents the

- A. frequency
- B. mean
- C. sector angle
- D. range

**40.** For a pictograph to be read correctly, it must include

- A. a key stating what each symbol represents
- B. sector angles
- C. equal bars
- D. a circle

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

(a) Data collection.

### **Question 2**

(a) Questionnaires.

### **Question 3**

(a) From questionnaire to graph.

### **Question 4**

(a) Reading a frequency table.

### **Question 5**

(a) Pie chart.

### **Question 6**

(a) Creating and reasoning.

# Chapter 7: Ch16

Time Allowed: 2 Hours | Total Marks: 100

## SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The mean of a data set is found by
  - A. taking the value that occurs most often
  - B. taking the middle value when the data is arranged in order
  - C. dividing the sum of all the values by the number of values
  - D. subtracting the smallest value from the largest
2. The mean of the data set {8, 9, 7, 6, 8, 10} is
  - A. 8
  - B. 6
  - C. 7
  - D. 9
3. The sum of five numbers is 60. The mean of the numbers is
  - A. 55
  - B. 300
  - C. 5
  - D. 12
4. The mean of 4, 8 and 6 is
  - A. 4
  - B. 6
  - C. 8
  - D. 18
5. The mean of five numbers is 10. The sum of the five numbers is
  - A. 50
  - B. 15
  - C. 2
  - D. 10
6. Ama scored 7, 8, 6 and 9 in four tests. Her mean score is
  - A. 7
  - B. 7.5
  - C. 8
  - D. 30
7. The mean of 2, 4, 6, 8 and 10 is
  - A. 5
  - B. 30
  - C. 6
  - D. 8
8. If every value in a data set is increased by 3, the mean of the data set
  - A. increases by 3
  - B. remains the same
  - C. increases by 6
  - D. becomes 3
9. The marks obtained by 20 students in a test are shown below: Mark 1 → 2 students, Mark 2 → 6 students, Mark 3 → 4 students, Mark 4 → 5 students, Mark 5 → 3 students. The mean mark is
  - A. 3
  - B. 2.8
  - C. 3.5
  - D. 3.05
10. The mean age of a group of children is 12 years and the sum of their ages is 96 years. How many children are in the group?
  - A. 12
  - B. 8
  - C. 96
  - D. 108
11. A baker sold 20, 25, 30, 24 and 26 loaves of bread on five successive

days. The mean number of loaves sold per day is

- A. 24
- B. 26
- C. 30
- D. 25

- 12.** The mean of four numbers is 15. Three of the numbers are 12, 16 and 14. The fourth number is

- A. 12
- B. 15
- C. 18

- D. 60
- 13.** Which measure of central tendency is most affected by a very large value (outlier) in the data?
- A. mean
  - B. median
  - C. mode
  - D. none of them
- 15.** Kofi's bus fares to school on four days were GH¢3, GH¢2, GH¢4 and GH¢3. The mean daily fare is
- A. GH¢2
  - B. GH¢2.50
  - C. GH¢4
  - D. GH¢3
- 17.** The median of a data set is
- A. the value that occurs most often
  - B. the middle value when the values are arranged in order
  - C. the sum of the values divided by 2
  - D. the largest value in the data set
- 19.** The median of 19, 29, 36, 15 and 20 is
- A. 15
  - B. 36
  - C. 20
  - D. 19
- 21.** The median of 5, 2, 9, 1 and 7 is
- A. 2
  - B. 5
  - C. 9
  - D. 7
- 23.** The median of 12, 15, 10 and 18 is
- A. 12
  - B. 13.5
  - C. 15
  - D. 14
- 14.** Which measure of central tendency uses every value in the data set when it is being calculated?
- A. mode
  - B. mean
  - C. median
  - D. range
- 16.** Sena's scores in five subjects are 75, 87, 90, 88 and 79. If she wants a mean score of 85 over six subjects, what must she score in the sixth subject?
- A. 91
  - B. 85
  - C. 90
  - D. 89
- 18.** The first step in finding the median of a data set is to
- A. add all the values
  - B. find the mode
  - C. draw a bar chart
  - D. arrange the values in ascending or descending order
- 20.** The median of the data set 8, 9, 7, 6, 8, 10 is
- A. 8
  - B. 8.5
  - C. 7
  - D. 9
- 22.** When a data set has an even number of values, the median is found by
- A. taking the larger of the two middle values
  - B. taking the smaller of the two middle values
  - C. taking the mean of the two middle values
  - D. adding all the values
- 24.** The heights, in cm, of six students are 120, 135, 118, 127, 140 and 130. The median height is
- A. 128.5
  - B. 127
  - C. 130
  - D. 128

- 25.** The table shows the marks obtained by 20 students: Mark 1 → 2 students, Mark 2 → 6 students, Mark 3 → 4 students, Mark 4 → 5 students, Mark 5 → 3 students. The median mark is
- 2
  - 3.05
  - 3
  - 4
- 27.** For a data set of 20 values arranged in order, the median is the mean of the
- 10th and 11th values
  - 9th and 10th values
  - 10th value only
  - 20th value only
- 29.** The mode of a data set is
- the middle value
  - the value that occurs most often
  - the sum of the values
  - the average of the values
- 31.** A football team scored the following goals in seven matches: 2, 0, 3, 1, 4, 2, 2. The mode of the goals is
- 0
  - 1
  - 2
  - 4
- 33.** The mode(s) of the data set 1, 2, 2, 3, 4, 4, 5 is/are
- 2 only
  - 4 only
  - 3
  - 2 and 4
- 35.** The table shows the marks obtained by 20 students: Mark 1 → 2 students, Mark 2 → 6 students, Mark 3 → 4 students, Mark 4 → 5 students, Mark 5 → 3 students. The modal mark is
- 2
  - 3
  - 4
  - 5
- 37.** Which measure of central tendency can be used for qualitative data such as favourite colours?
- mean
- 26.** The values 4, 6, 7, 9 and 11 are arranged in order. If the greatest value is changed from 11 to 20, the median
- increases to 20
  - becomes 8
  - decreases
  - remains 7
- 28.** The median of 2, 2, 2, 2 and 10 is 2. This shows that the median
- uses all the values
  - is strongly affected by the outlier 10
  - is not affected by the extreme value 10
  - always equals the mean
- 30.** The mode of 2, 3, 3, 5, 3, 6 is
- 2
  - 5
  - 6
  - 3
- 32.** A data set in which no value occurs more than once has
- no mode
  - a mode of 0
  - every value as its mode
  - the mean as its mode
- 34.** The best measure of central tendency to use when finding the most popular drink at a party is the
- mean
  - mode
  - median
  - range
- 36.** The mean of six numbers is 7. Five of the numbers are 5, 8, 6, 9 and 7. The sixth number is
- 6
  - 35
  - 7
  - 8

- B. median
- C. range
- D. mode

**39.** The median of 4, 6, 6 and 8 is

- A. 4
- B. 8
- C. 6
- D. 7

**38.** The mode of the data 5, 8, 6, 9, 7, 9 is

- A. 8
- B. 9
- C. 7.5
- D. 6

- 40.** For the data set 4, 6, 6, 8, which of the following statements is true?
- A. mean = median = mode
  - B. mean is greater than the mode
  - C. the median is 8
  - D. the mode is 4

**SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

**Question 1**

- (a) The mean.

**Question 2**

- (a) The median.

**Question 3**

- (a) The mode.

**Question 4**

- (a) Central tendency from a frequency table.

**Question 5**

- (a) Solving problems with the mean.

**Question 6**

- (a) Creating and reasoning.

## Chapter 8: Ch1

Time Allowed: 2 Hours | Total Marks: 100

### SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Probability is a measure of
  - A. how fast an event happens
  - B. how likely an event is to happen
  - C. the size of an event
  - D. the number of events that happen
2. "The dog will fly tomorrow." This event is
  - A. certain
  - B. possible
  - C. impossible
  - D. equally likely
3. "Ghana will still be an African country tomorrow." This event is
  - A. certain
  - B. impossible
  - C. doubtful
  - D. possible
4. "Someone in this class will be a teacher in the future." This event is
  - A. impossible
  - B. possible
  - C. certain
  - D. sure
5. An event that will definitely happen is described as
  - A. an event that can never happen
  - B. an event that may or may not happen
  - C. certain
  - D. an event with probability 0
6. An event that can never happen is described as
  - A. impossible
  - B. certain
  - C. an event with probability 1
  - D. possible
7. "The day after Monday will be Tuesday." This event is
  - A. possible
  - B. impossible
  - C. certain
  - D. uncertain
8. "A coin lands heads side up." This event is
  - A. certain
  - B. possible
  - C. impossible
  - D. sure
9. A fair die is thrown once. Obtaining the number 7 is
  - A. possible
  - B. certain
  - C. impossible
  - D. equally likely
10. Two fair dice are thrown. Obtaining a total of 13 is
  - A. impossible
  - B. possible
  - C. certain
  - D. equally likely
11. Two fair dice are thrown. Obtaining a total of 12 is
  - A. impossible
  - B. certain
  - C. possible
  - D. equally likely
12. "It will rain in Winneba in the first week of January." This event is
  - A. impossible
  - B. possible
  - C. certain
  - D. sure

- 13.** The sample space of a probability experiment is
- A. the outcomes that occur most often
  - B. only one outcome of the experiment
  - C. the set of all possible outcomes
  - D. the outcomes that cannot happen
- 15.** How many outcomes are there when a fair die is thrown once?
- A. 6
  - B. 5
  - C. 12
  - D. 2
- 17.** Two fair coins are tossed together. The number of possible outcomes is
- A. 2
  - B. 3
  - C. 6
  - D. 4
- 19.** A bag contains 3 red and 2 blue counters. One counter is picked at random. The sample space is
- A. {3, 2}
  - B. {red, blue}
  - C. {red}
  - D. {5}
- 21.** A fair die is thrown once. The outcomes that make up the event "a number greater than 4" are
- A. {4, 5, 6}
  - B. {5, 6}
  - C. {6}
  - D. {1, 2, 3, 4}
- 23.** The probability of an event is calculated as
- A.  $\text{total outcomes} \div \text{favourable outcomes}$
  - B.  $\text{favourable outcomes} \div \text{total outcomes}$
  - C.  $\text{favourable outcomes} + \text{total outcomes}$
  - D.  $\text{favourable outcomes} \times \text{total outcomes}$
- 25.** A fair die is thrown once. The probability of obtaining an even number is
- 14.** The sample space for tossing one fair coin is
- A. {H}
  - B. {H, T}
  - C. {H, T, S}
  - D. {T}
- 16.** The sample space for throwing one fair die is
- A. {1, 2, 3, 4, 5}
  - B. {2, 4, 6}
  - C. {1, 2, 3, 4, 5, 6}
  - D. {1, 3, 5}
- 18.** Which of the following is the sample space for tossing two fair coins?
- A. HH, HT, TH, TT
  - B. HH, TT only
  - C. H, T only
  - D. HT, TH only
- 20.** A spinner is divided into 5 equal sectors. How many possible outcomes does it have?
- A. 1
  - B. 10
  - C. 25
  - D. 5
- 22.** How many even numbers are in the sample space of a fair die?
- A. 2
  - B. 4
  - C. 3
  - D. 6
- 24.** A fair die is thrown once. The probability of obtaining a 4 is
- A.  $\frac{1}{6}$
  - B.  $\frac{4}{6}$
  - C.  $\frac{1}{4}$
  - D. 4

- A.  $\frac{1}{3}$
- B.  $\frac{2}{3}$
- C.  $\frac{1}{6}$
- D.  $\frac{1}{2}$

**27.** A fair die is thrown once. The probability of obtaining a factor of 6 is

- A.  $\frac{1}{2}$
- B.  $\frac{1}{3}$
- C. 1
- D.  $\frac{2}{3}$

**26.** A fair die is thrown once. The probability of obtaining a multiple of 3 is

- A.  $\frac{1}{2}$
- B.  $\frac{1}{6}$
- C.  $\frac{1}{3}$
- D.  $\frac{2}{3}$

- 29.** A fair die is thrown once. The probability of obtaining a divisor of 30 is
- A.  $\frac{5}{6}$
  - B.  $\frac{2}{3}$
  - C.  $\frac{1}{2}$
  - D. 1
- 31.** A fair die is thrown once. The probability of obtaining a 3 or smaller is
- A.  $\frac{1}{2}$
  - B.  $\frac{2}{3}$
  - C.  $\frac{1}{3}$
  - D.  $\frac{1}{6}$
- 33.** The probability of a certain event is
- A. 0
  - B.  $\frac{1}{2}$
  - C. 1
  - D. 10
- 35.** Express the probability  $\frac{1}{4}$  as a decimal.
- A. 0.25
  - B. 0.14
  - C. 0.4
  - D. 0.75
- 37.** A probability of  $\frac{1}{2}$  expressed as a ratio is
- A. 1 : 2
  - B. 2 : 1
  - C. 1 : 1
  - D. 2 : 3
- 39.** A spinner has 4 equal sectors, one of which is red. The probability that the spinner lands on red is
- A.  $\frac{1}{2}$
  - B.  $\frac{1}{4}$
  - C.  $\frac{3}{4}$
  - D.  $\frac{1}{5}$
- 28.** A fair die is thrown once. The probability of obtaining a factor of 8 is
- A.  $\frac{1}{3}$
  - B.  $\frac{1}{2}$
  - C.  $\frac{2}{3}$
  - D.  $\frac{1}{6}$
- 30.** A fair die is thrown once. The probability of obtaining a 3 or greater is
- A.  $\frac{1}{2}$
  - B.  $\frac{1}{3}$
  - C.  $\frac{5}{6}$
  - D.  $\frac{2}{3}$
- 32.** The probability of an impossible event is
- A. 1
  - B. 0
  - C.  $\frac{1}{2}$
  - D. 100
- 34.** A bag contains 3 red and 2 blue counters. One counter is picked at random. The probability that it is red is
- A.  $\frac{2}{5}$
  - B.  $\frac{1}{5}$
  - C.  $\frac{3}{2}$
  - D.  $\frac{3}{5}$
- 36.** The probability of an event is  $\frac{1}{2}$ . Expressed as a percentage, it is
- A. 25%
  - B. 12%
  - C. 20%
  - D. 50%
- 38.** A fair coin is tossed once. The probability of obtaining heads is
- A.  $\frac{1}{4}$
  - B.  $\frac{1}{3}$
  - C.  $\frac{1}{2}$
  - D. 1
- 40.** A bag contains 2 mangoes, 3 oranges and 5 bananas. One fruit is picked at random. The probability that it is an orange is
- A.  $\frac{3}{5}$
  - B.  $\frac{1}{2}$
  - C.  $\frac{2}{10}$
  - D.  $\frac{3}{10}$

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

(a) Classifying likelihood.

### **Question 2**

(a) Sample space.

### **Question 3**

(a) Calculating probability with a die.

### **Question 4**

(a) Expressing probabilities in different forms.

### **Question 5**

(a) Probability from a bag.

### **Question 6**

(a) Creating and reasoning.

# Chapter 9: Ch1

Time Allowed: 2 Hours | Total Marks: 100

## SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. What is the place value of the digit 7 in the number 4,275,308,619?
  - A. Millions
  - B. Ten millions
  - C. Hundred millions
  - D. Ten billions
2. Which of the following is six billion, three hundred and twenty million, four hundred and fifty thousand, eight hundred and seven written in numerals?
  - A. 6,302,405,807
  - B. 6,302,450,807
  - C. 6,320,450,807
  - D. 6,320,450,870
3. In the number 5,432,167,890, what is the value of the digit 4?
  - A. 40,000,000
  - B. 400,000,000
  - C. 4,000,000,000
  - D. 4,000,000
4. How many millions make one billion?
  - A. 10
  - B. 100
  - C. 1,000
  - D. 10,000
5. The Government of Ghana budgeted GH¢12,406,000,000 for education in the year 2026/27. How is this amount written in words?
  - A. Twelve billion, forty and six million cedis
  - B. Twelve billion, four hundred and sixty million cedis
  - C. One hundred and twenty-four billion, six million cedis
  - D. Twelve billion, four hundred and six million cedis
6. Which digit occupies the hundred millions place in the number 9,807,654,321?
  - A. 9
  - B. 8
  - C. 0
  - D. 7
7. Which of the following is the expanded form of 3,045,000,200?
  - A.  $3,000,000,000 + 45,000,000 + 200$
  - B.  $3,000,000,000 + 4,500,000 + 200$
  - C.  $3,000,000,000 + 45,000,000 + 2,000$
  - D.  $300,000,000 + 45,000,000 + 200$
8. What is the sum of the value of the digit 6 and the value of the digit 2 in the number 2,615,000,000?
  - A. 2,060,000,000
  - B. 260,000,000
  - C. 2,600,000,000
  - D. 8
9. Round 7,823,456,000 to the nearest hundred thousand.
  - A. 7,823,500,000
  - B. 7,823,400,000
  - C. 7,820,000,000
  - D. 7,823,460,000
10. Round 2,099,964,000 to the nearest ten thousand.
  - A. 2,099,970,000
  - B. 2,100,000,000
  - C. 2,099,960,000
  - D. 2,099,900,000

- 11.** Round 5,600,349,200 to the nearest thousand.
- A. 5,600,350,000
  - B. 5,600,340,000
  - C. 5,600,400,000
  - D. 5,600,349,000
- 12.** Round 3,714,082,547 to the nearest hundred.
- A. 3,714,082,600
  - B. 3,714,082,500
  - C. 3,714,083,000
  - D. 3,714,080,000
- 13.** Round 1,258,437,963 to the nearest ten.
- A. 1,258,437,960
  - B. 1,258,437,970
  - C. 1,258,438,000
  - D. 1,258,437,900
- 14.** A number rounded to the nearest hundred thousand gives 2,400,000,000. Which of the following could be the number?
- A. 2,399,949,000
  - B. 2,400,051,000
  - C. 2,400,049,000
  - D. 2,399,850,000
- 15.** Round 6.74 to the nearest tenth.
- A. 6.8
  - B. 7.0
  - C. 6.0
  - D. 6.7
- 16.** Round 15.096 to the nearest hundredth.
- A. 15.10
  - B. 15.09
  - C. 15.11
  - D. 15.96
- 17.** Round 0.997 to the nearest hundredth.
- A. 0.99
  - B. 1.00
  - C. 0.97
  - D. 1.10
- 18.** Round 243.6584 to the nearest thousandth.
- A. 243.659
  - B. 243.66
  - C. 243.658
  - D. 243.7
- 19.** Round down 3,875,240,000 to the nearest million.
- A. 3,876,000,000
  - B. 3,870,000,000
  - C. 3,800,000,000
  - D. 3,875,000,000
- 20.** The total mobile money transactions in a region of Ghana in 2025 was GH¢2,847,516,000. What is this amount to the nearest million cedis?
- A. GH¢2,848,000,000
  - B. GH¢2,847,000,000
  - C. GH¢2,800,000,000
  - D. GH¢2,850,000,000
- 21.** How many decimal places does the number 0.0405 have?
- A. 2
  - B. 4
  - C. 3
  - D. 5
- 22.** Express 3.14159 correct to 3 decimal places.
- A. 3.141
  - B. 3.14
  - C. 3.1415
  - D. 3.142
- 23.** Express 0.0568 correct to 2 significant figures.
- A. 0.056
  - B. 0.06
  - C. 0.057
  - D. 0.0568
- 24.** Express 45,932,000 correct to 3 significant figures.
- A. 45,930,000
  - B. 45,900,000
  - C. 46,000,000
  - D. 45,000,000
- 25.** How many significant figures are there in the number 6.030?
- A. 4
  - B. 3
  - C. 2
  - D. 5

- 27.** Which symbol makes the statement true?  $3,205,000,000 \square 3,200,500,000$
- $>$
  - $<$
  - $=$
  - $\times$
- 29.** Which of the following numbers is the greatest?
- 2,099,999,999
  - 2,009,999,999
  - 2,100,000,000
  - 2,090,999,999
- 31.** Which symbol is used to mean “less than”?
- $>$
  - $<$
  - $=$
  - $\neq$
- 33.** When the number  $3,2\square 6,000,000$  is rounded to the nearest ten million, the result is  $3,300,000,000$ . What digit does  $\square$  represent?
- 6
  - 7
  - 8
  - 9
- 26.** In the year 2026, the population of the world was estimated at 8,164,000,000. Express this population correct to 1 significant figure.
- 8,100,000,000
  - 8,160,000,000
  - 800,000,000
  - 8,000,000,000
- 28.** Arrange the following numbers in ascending order: 1,250,000,000; 1,205,000,000; 1,502,000,000; 1,052,000,000.
- 1,250,000,000; 1,205,000,000; 1,502,000,000; 1,052,000,000
  - 1,052,000,000; 1,205,000,000; 1,250,000,000; 1,502,000,000
  - 1,052,000,000; 1,250,000,000; 1,205,000,000; 1,502,000,000
  - 1,502,000,000; 1,250,000,000; 1,205,000,000; 1,052,000,000
- 30.** Four mobile network companies recorded the following data usage in a year: MTN 5,107,020,000 MB; Telecel 3,910,450,000 MB; AirtelTigo 4,012,300,000 MB; Expresso 3,999,990,000 MB. Arrange the companies in order from the least to the greatest data usage.
- Expresso, Telecel, AirtelTigo, MTN
  - Telecel, AirtelTigo, Expresso, MTN
  - MTN, AirtelTigo, Expresso, Telecel
  - Telecel, Expresso, AirtelTigo, MTN
- 32.** What is the value of  $7,000,000,000 + 40,000,000 + 90,000 + 600$ ?
- 7,400,090,600
  - 7,004,090,600
  - 7,040,090,600
  - 7,040,900,006
- 34.** Abena’s family business recorded sales of GH¢1,075,499,996 in 2025. What is this amount to the nearest hundred thousand cedis?
- GH¢1,075,400,000
  - GH¢1,075,500,000

- C. GH¢1,076,000,000
- D. GH¢1,070,000,000

**36.** Express 2.0060 correct to 3 significant figures.

- A. 2.00
- B. 2.006
- C. 2.1
- D. 2.01

**38.** The population of the world is about eight billion, one hundred and sixty million. Which of the following is this number written in numerals?

- A. 8,016,000,000
- B. 8,160,000,000
- C. 816,000,000
- D. 8,106,000,000

**35.** Which of the following numbers becomes 5,000,000,000 when rounded to the nearest hundred million?

- A. 4,950,000,000
- B. 4,949,000,000
- C. 5,050,000,000
- D. 4,850,000,000

**37.** I am a ten-digit number. My billions digit is 4. My hundred millions digit is half of my billions digit. My ten millions digit is three times my hundred millions digit. All my other digits are zeros. What number am I?

- A. 4,620,000,000
- B. 2,460,000,000
- C. 4,260,000,000
- D. 4,206,000,000

**39.** Round 6.996 correct to 2 decimal places.

- A. 7.00

- B. 6.99
- C. 6.96
- D. 7.10

- 40.** Esi was asked to round 3,450,000,000 to the nearest hundred thousand and she answered 3,400,000,000. Which of the following statements is true?
- A. Esi is correct because the number ends in zeros.
  - B. Esi is correct because 5 must always be rounded down.
  - C. Esi is wrong; the correct answer is 3,450,000,000.
  - D. Esi is wrong; the correct answer is 3,500,000,000.

## SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

### Question 1

- (a) A mobile network company sent a total of 4,073,605,218 text messages in one year.

### Question 2

- (a) Four companies in Ghana recorded the following sales in the 2025/26 financial year:

### Question 5

- (a) The Government of Ghana budgeted GH¢12,406,385,000 for the construction of schools in the 2026/27 academic year.

### Question 6

- (a) Adjoa was given the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9 and asked to use each digit once to form ten-digit numbers.

# Chapter 10: Ch2

Time Allowed: 2 Hours | Total Marks: 100

## SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. What is  $3.87 \times 100$ ?  
A. 38.7  
B. 387  
C. 3,870  
D. 0.0387
2. What is  $65.4 \div 10$ ?  
A. 654  
B. 0.654  
C. 6.54  
D. 6,540
3. What is  $0.075 \times 1,000$ ?  
A. 7.5  
B. 750  
C. 0.75  
D. 75
4. What is  $902 \div 100$ ?  
A. 9.02  
B. 90.2  
C. 0.902  
D. 90,200
5. Evaluate  $4.6 \times 10$ .  
A. 0.46  
B. 460  
C. 46  
D. 4.6
6. What is  $7/10 \times 100$ ?  
A. 7  
B. 70  
C. 700  
D. 0.7
7. What is  $3/4 \times 1,000$ ?  
A. 75  
B. 250  
C. 3,400  
D. 750
8. The statement  $17 + 26 = 26 + 17$  illustrates the  
A. associative property of addition  
B. commutative property of addition  
C. distributive property of multiplication  
D. identity property of addition
9. To evaluate  $25 \times 19$  as  $25 \times (20 - 1)$ , which property of numbers is used?  
A. Commutative property  
B. Associative property  
C. Distributive property  
D. Zero property
10. Using mental mathematics,  $12 \times 99$  =  
A. 1,188  
B. 1,089  
C. 1,200  
D. 1,098
11. What is  $0 \times 4,387$ ?  
A. 4,387  
B. 1  
C. 43,870  
D. 0
12. The statement  $36 + 0 = 36$  illustrates the  
A. commutative property  
B. distributive property  
C. identity property of addition  
D. associative property
13. Using mental mathematics,  $48 \times 5$  =  
A. 240  
B. 2,400  
C. 204  
D. 480

- 15.** Evaluate  $4 \times 17 \times 25$  mentally by choosing a convenient order.
- A. 170
  - B. 1,070
  - C. 17,000
  - D. 1,700
- 17.** Amina thinks of a number. When she multiplies it by 10, she gets 85. What is the number?
- A. 850
  - B. 8.5
  - C. 0.85
  - D. 85
- 19.** Evaluate  $3,758 + 2,467$ .
- A. 6,225
  - B. 6,215
  - C. 5,225
  - D. 6,125
- 21.** A shop sold 4,386 bottles of water in January and 2,947 bottles in February. How many bottles were sold altogether?
- A. 7,323
  - B. 7,333
  - C. 6,333
  - D. 7,233
- 23.** Find the missing number:  $2,415 + \square = 6,000$ .
- A. 3,585
  - B. 3,685
  - C. 4,585
  - D. 3,595
- 25.** Evaluate  $436 \times 7$ .
- A. 3,042
  - B. 3,052
  - C. 2,952
  - D. 3,152
- 27.** Evaluate  $852 \div 6$ .
- A. 124
  - B. 148
  - C. 1,420
  - D. 142
- 29.** Evaluate  $6,048 \div 24$ .
- A. 225
  - B. 252
  - C. 262
  - D. 2,520
- 14.** Using compensation,  $76 + 58 =$
- A. 124
  - B. 134
  - C. 144
  - D. 136
- 16.** Kwame has GH¢50.00. He buys a book for GH¢28.50. What is his change, worked out mentally?
- A. GH¢22.50
  - B. GH¢21.05
  - C. GH¢21.50
  - D. GH¢78.50
- 18.** The fare from Accra to Kasoa is GH¢12.00. What is the total fare for 25 passengers, worked out mentally?
- A. GH¢250.00
  - B. GH¢325.00
  - C. GH¢300.00
  - D. GH¢3,000.00
- 20.** Evaluate  $6,004 - 2,587$ .
- A. 4,417
  - B. 3,427
  - C. 3,517
  - D. 3,417
- 22.** Subtract 1,896 from 5,210.
- A. 4,314
  - B. 3,324
  - C. 3,686
  - D. 3,314
- 24.** What is the difference between 7,031 and 2,645?
- A. 4,486
  - B. 5,386
  - C. 4,386
  - D. 4,396
- 26.** Evaluate  $285 \times 34$ .
- A. 9,690
  - B. 9,590
  - C. 8,690
  - D. 9,780
- 28.** Evaluate  $1,275 \div 15$ .
- A. 75
  - B. 95
  - C. 85
  - D. 805

- 31.** Evaluate  $306 \times 25$ .
- A. 7,560
  - B. 6,750
  - C. 76,500
  - D. 7,650
- 33.** Mrs Adjei bought 14.8 kg of fish. Mrs Mensah bought 3.2 kg less than Mrs Adjei. How many kilograms of fish did they buy altogether?
- A. 26.4 kg
  - B. 18.0 kg
  - C. 11.6 kg
  - D. 29.6 kg
- 35.** Mrs Owusu bought 52.25 m of wax print and shared it equally among her five children. How many metres did each child receive?
- A. 10.54 m
  - B. 10.45 m
  - C. 9.45 m
  - D. 10.25 m
- 37.** One bag of maize weighs 25.6 kg. What is the total weight of 3 such bags?
- A. 75.8 kg
  - B. 78.6 kg
  - C. 76.8 kg
  - D. 28.6 kg
- 39.** Evaluate  $8.64 \div 4$ .
- A. 21.6
  - B. 2.16
  - C. 2.61
  - D. 0.216
- 30.** When 947 is divided by 8, what are the quotient and the remainder?
- A. 118 remainder 3
  - B. 117 remainder 7
  - C. 118 remainder 5
  - D. 119 remainder 1
- 32.** A farmer packs 1,440 eggs equally into boxes of 12. How many boxes does he need?
- A. 12
  - B. 1,200
  - C. 120
  - D. 102
- 34.** Kofi weighs 32.5 kg. His father weighs four times as much as Kofi. What is the total weight of Kofi and his father?
- A. 130 kg
  - B. 162 kg
  - C. 135.5 kg
  - D. 162.5 kg
- 36.** Efe had GH¢20.00. She bought sugar for GH¢7.35 and milk for GH¢5.80. How much money was left?
- A. GH¢6.85
  - B. GH¢7.15
  - C. GH¢13.15
  - D. GH¢6.58
- 38.** A driver bought 45.5 litres of fuel and used 18.7 litres. How much fuel is left?
- A. 27.8 litres
  - B. 26.2 litres
  - C. 64.2 litres
  - D. 26.8 litres
- 40.** Kojo ran 3.75 km on Saturday and 4.6 km on Sunday. How far did he run altogether?
- A. 7.35 km
  - B. 8.25 km
  - C. 8.35 km
  - D. 83.5 km

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

(a) Powers of 10.

### **Question 2**

(a) Number properties and mental mathematics.

### **Question 3**

(a) Addition and subtraction of whole numbers.

### **Question 4**

(a) Multiplication and division of multi-digit numbers.

### **Question 5**

(a) Story problems involving decimals.

### **Question 6**

(a) Creating and solving decimal story problems.

# Chapter 11: Ch3

Time Allowed: 2 Hours | Total Marks: 100

## SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Express  $3 \times 3 \times 3 \times 3$  in index form.  
A.  $3 \times 4$   
B.  $4^3$   
C.  $3^4$   
D. 12
2. The expression  $2^5$  means  
A.  $2 \times 5$   
B.  $2 + 2 + 2 + 2 + 2$   
C.  $5 \times 5$   
D.  $2 \times 2 \times 2 \times 2 \times 2$
3. Which of the following represents the repeated multiplication  $6 \times 6 \times 6$ ?  
A.  $6^3$   
B.  $3^6$   
C.  $6 \times 3$   
D. 18
4. Kofi arranges bottle tops into 5 heaps of 5 caps each. The total number of caps is  $5 \times 5$ . How is  $5 \times 5$  written in index form?  
A.  $5^3$   
B.  $2^5$   
C.  $5^2$   
D.  $5 \times 2$
5. How many times is 7 used as a factor in the expression  $7^6$ ?  
A. 7  
B. 6  
C. 13  
D. 42
6. Which of the following represents  $2 \times 2 \times 2 \times 2 \times 2$ ?  
A.  $6^2$   
B. 12  
C.  $6 \times 2$   
D.  $2^6$
7. In the expression  $9^4$ , the base and the exponent are respectively  
A. 4 and 9  
B. 9 and 4  
C. 9 and 36  
D. 13 and 9
8. Express 64 as a power of 2.  
A.  $2^4$   
B.  $2^5$   
C.  $2^6$   
D.  $2^8$
9. Express 81 as a power of 3.  
A.  $3^4$   
B.  $3^3$   
C.  $4^3$   
D.  $9^4$
10. What is the value of  $2^3 \times 3^2$ ?  
A. 36  
B. 24  
C. 12  
D. 72
11. Which of the following expresses 100 as a product of powers of its prime factors?  
A.  $2 \times 5^2$   
B.  $2^2 \times 5$   
C.  $2^2 \times 5^2$   
D.  $2^5$
12. Write  $5 \times 5 \times 5 \times 2 \times 2$  in index form.  
A.  $5^2 \times 2^3$   
B.  $5^3 \times 2^2$   
C.  $10^5$   
D.  $5 \times 2$

**13.** Express 1,000 as a power of 10.

- A.  $10^2$
- B.  $10^4$
- C.  $3^{10}$
- D.  $10^3$

**15.** Which of the following is equal to  $4 \times 4 \times 4 \times 4$ ?

- A.  $4^4$
- B. 16
- C.  $4 \times 4$
- D.  $8^2$

**17.** Which of the following statements is true?

- A. Every natural number raised to the power 0 gives 0
- B.  $5^0 = 5$
- C.  $0^5 = 1$
- D. Every natural number raised to the power 0 gives 1

**19.** Evaluate  $4 \times 6^0$ .

- A. 24
- B. 0
- C. 4
- D. 10

**21.** Find the value of  $2^5$ .

- A. 10
- B. 25
- C. 16
- D. 32

**23.** Find the value of  $10^4$ .

- A. 10,000
- B. 40
- C. 1,000
- D. 100,000

**25.** Find the value of  $6^3$ .

- A. 18
- B. 216
- C. 36
- D. 126

**27.** Evaluate  $2^3 + 3^2$ .

- A. 13
- B. 25
- C. 17
- D. 36

**29.** Evaluate  $(2 \times 3)^2$ .

- A. 12

**14.** What is the value of  $7^2 \times 2^3$ ?

- A. 98
- B. 392
- C. 343
- D. 56

**16.** What is the value of  $9^0$ ?

- A. 0
- B. 1
- C. 9
- D. 90

**18.** Evaluate  $7^0 + 3^0$ .

- A. 2
- B. 10
- C. 1
- D. 0

**20.** If  $5^n = 1$ , then the value of n is

- A. 1
- B. 0
- C. 5
- D. 10

**22.** Find the value of  $5^3$ .

- A. 15
- B. 125
- C. 25
- D. 53

**24.** Find the value of  $3^4$ .

- A. 12
- B. 27
- C. 64
- D. 81

**26.** Which of the following numbers is a perfect square?

- A. 64
- B. 48
- C. 50
- D. 72

**28.** Evaluate  $5^2 - 4^2$ .

- A. 1
- B. 9
- C. 20
- D. 41

- B. 18
- C. 23
- D. 36

**30.** Which of the following is equal to 125?

- A.  $5^3$
- B.  $3^5$
- C.  $25^2$

- D.  $5^2$
- 31.** Evaluate  $2^0 + 2^1 + 2^2 + 2^3$ .  
A. 8  
B. 14  
C. 15  
D. 16
- 33.** Express 36 as a product of powers of its prime factors.  
A.  $2 \times 3^2$   
B.  $2^2 \times 3^2$   
C.  $2^3 \times 3$   
D.  $4 \times 9$
- 35.** Find the HCF of 24 and 36.  
A. 6  
B. 18  
C. 12  
D. 24
- 37.** Find the HCF of  $2^3 \times 3^2$  and  $2^2 \times 3^3$ .  
A.  $2 \times 3$   
B.  $2^2 \times 3^2$   
C.  $2^3 \times 3^3$   
D.  $2^2 \times 3^3$
- 39.** Find the HCF of 16 and 48.  
A. 4  
B. 8  
C. 32  
D. 16
- 32.** If  $3^n = 81$ , then the value of n is  
A. 3  
B. 9  
C. 27  
D. 4
- 34.** Find the HCF of 12 and 18.  
A. 6  
B. 2  
C. 3  
D. 36
- 36.** Find the HCF of 24 and 60.  
A. 4  
B. 8  
C. 6  
D. 12
- 38.** Find the HCF of 45 and 75.  
A. 15  
B. 5  
C. 3  
D. 225
- 40.** A teacher has 48 pencils and 36 rulers. She wants to make identical packs so that no item is left over. What is the greatest number of packs she can make?  
A. 6  
B. 144  
C. 12  
D. 18

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

- (a) Repeated factors and index form.

### **Question 2**

- (a) Expressing numbers as products of powers.

### **Question 3**

- (a) The zero exponent.

### **Question 4**

- (a) Finding the value of numbers in index form.

### **Question 5**

- (a) Highest Common Factor by the product of prime factors.

### **Question 6**

- (a) Integrated problems and reasoning.

## Chapter 12: Ch

Time Allowed: 2 Hours | Total Marks: 100

### SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Express  $\frac{1}{2}$  as a percentage.
  - A. 5%
  - B. 2%
  - C. 50%
  - D. 20%
2. Express  $\frac{3}{4}$  as a decimal.
  - A. 0.34
  - B. 0.75
  - C. 7.5
  - D. 3.4
3. Which fraction in its lowest terms is equal to 20%?
  - A.  $\frac{1}{5}$
  - B.  $\frac{1}{20}$
  - C.  $\frac{2}{5}$
  - D.  $\frac{5}{2}$
4. Which of the following fractions is equivalent to 0.4?
  - A.  $\frac{1}{4}$
  - B.  $\frac{4}{5}$
  - C.  $\frac{4}{50}$
  - D.  $\frac{2}{5}$
5. Which of the following is  $\frac{1}{3}$  written as a decimal?
  - A. 0.30
  - B. 3.0
  - C. 0.333...
  - D. 0.033
6. Express 40% as a decimal.
  - A. 0.4
  - B. 0.04
  - C. 4.0
  - D. 0.14
7. Which of the following fractions is greater?
  - A.  $\frac{2}{3}$
  - B.  $\frac{3}{4}$
  - C. They are equal
  - D. It cannot be determined
8. Arrange the fractions  $\frac{5}{6}$ ,  $\frac{3}{4}$  and  $\frac{7}{8}$  in descending order.
  - A.  $\frac{3}{4}$ ,  $\frac{5}{6}$ ,  $\frac{7}{8}$
  - B.  $\frac{5}{6}$ ,  $\frac{7}{8}$ ,  $\frac{3}{4}$
  - C.  $\frac{7}{8}$ ,  $\frac{3}{4}$ ,  $\frac{5}{6}$
  - D.  $\frac{7}{8}$ ,  $\frac{5}{6}$ ,  $\frac{3}{4}$
9. Which of the following decimal fractions is greater?
  - A. 0.977
  - B. 0.99 and 0.977 are equal
  - C. 0.99
  - D. It cannot be determined
10. Arrange the decimals 0.098, 0.985 and 0.123 in ascending order.
  - A. 0.098, 0.123, 0.985
  - B. 0.123, 0.098, 0.985
  - C. 0.098, 0.985, 0.123
  - D. 0.985, 0.123, 0.098
11. Which of the following is equivalent to  $\frac{3}{5}$ ?
  - A. 0.35
  - B. 30%
  - C.  $\frac{5}{3}$
  - D. 0.6
12. Which of the following fractions is the least?
  - A.  $\frac{2}{5}$
  - B.  $\frac{1}{3}$
  - C.  $\frac{3}{8}$
  - D.  $\frac{1}{2}$

- 13.** Arrange 25%,  $\frac{1}{3}$  and 0.3 in ascending order.  
A. 25%, 0.3,  $\frac{1}{3}$   
B. 0.3, 25%,  $\frac{1}{3}$   
C.  $\frac{1}{3}$ , 0.3, 25%  
D. 25%,  $\frac{1}{3}$ , 0.3
- 15.** Evaluate  $\frac{1}{4} + \frac{2}{4}$ .  
A.  $\frac{3}{8}$   
B.  $\frac{3}{4}$   
C.  $\frac{2}{16}$   
D.  $\frac{1}{2}$
- 17.** Evaluate  $\frac{2}{3} + \frac{1}{4}$ .  
A.  $\frac{11}{12}$   
B.  $\frac{3}{7}$   
C.  $\frac{3}{12}$   
D.  $\frac{8}{12}$
- 19.** Evaluate  $2\frac{2}{5} + 1\frac{2}{3}$ .  
A.  $3\frac{4}{15}$   
B.  $4\frac{1}{15}$   
C.  $3\frac{16}{15}$   
D.  $4\frac{4}{15}$
- 21.** Evaluate  $3\frac{1}{2} - 1\frac{3}{4}$ .  
A.  $2\frac{1}{2}$   
B.  $1\frac{1}{4}$   
C.  $2\frac{1}{4}$   
D.  $1\frac{3}{4}$
- 23.** The Musa family hiked  $2\frac{1}{2}$  km in the morning and  $3\frac{3}{4}$  km in the afternoon. What was the total distance they hiked?  
A.  $6\frac{3}{8}$  km  
B.  $6\frac{1}{4}$  km  
C.  $5\frac{3}{4}$  km  
D.  $6\frac{1}{2}$  km
- 25.** Esi spent  $\frac{1}{2}$  of her money on rice and  $\frac{1}{4}$  of it on cooking oil. What fraction of her money is left?  
A.  $\frac{3}{4}$   
B.  $\frac{1}{2}$   
C.  $\frac{1}{8}$   
D.  $\frac{1}{4}$
- 27.** Find  $\frac{2}{3}$  of GH¢60.00.  
A. GH¢30.00  
B. GH¢40.00  
C. GH¢20.00  
D. GH¢90.00
- 14.** Which percentage is equal to  $\frac{7}{10}$ ?  
A. 7%  
B. 17%  
C. 70%  
D. 0.7%
- 16.** Evaluate  $\frac{1}{2} + \frac{1}{3}$ .  
A.  $\frac{2}{5}$   
B.  $\frac{2}{6}$   
C.  $\frac{1}{6}$   
D.  $\frac{5}{6}$
- 18.** Evaluate  $\frac{3}{4} - \frac{1}{3}$ .  
A.  $\frac{2}{1}$   
B.  $\frac{2}{7}$   
C.  $\frac{5}{12}$   
D.  $\frac{1}{6}$
- 20.** Evaluate  $5 - 2\frac{3}{4}$ .  
A.  $2\frac{1}{4}$   
B.  $2\frac{3}{4}$   
C.  $3\frac{1}{4}$   
D.  $2\frac{1}{2}$
- 22.** A plank is  $12\frac{1}{4}$  metres long. A carpenter cuts off  $3\frac{1}{3}$  metres. How long is the remaining part?  
A.  $9\frac{1}{12}$  m  
B.  $8\frac{1}{4}$  m  
C.  $8\frac{11}{12}$  m  
D.  $9\frac{1}{4}$  m
- 24.** Ama drank  $\frac{1}{3}$  of a bottle of water and Adjoa drank  $\frac{1}{4}$  of it. What fraction of the water did they drink altogether?  
A.  $\frac{7}{12}$   
B.  $\frac{2}{7}$   
C.  $\frac{1}{7}$   
D.  $\frac{2}{12}$
- 26.** Evaluate  $3 \times 2\frac{2}{3}$ .  
A.  $6\frac{2}{3}$   
B.  $7\frac{2}{3}$   
C. 8  
D. 24
- 28.** What is  $\frac{1}{4}$  of 100?  
A. 25  
B. 40  
C. 14  
D. 400

- 29.** Evaluate  $\frac{3}{5} \times 20$ .  
A. 15  
B. 30  
C. 3  
D. 12
- 31.** In a class of 40 pupils,  $\frac{5}{8}$  are girls.  
How many girls are in the class?  
A. 20  
B. 25  
C. 16  
D. 32
- 33.** Find 25% of GH¢80.00.  
A. GH¢40.00  
B. GH¢25.00  
C. GH¢8.00  
D. GH¢20.00
- 35.** Evaluate  $\frac{3}{4} \div 3$ .  
A.  $\frac{9}{4}$   
B.  $\frac{1}{4}$   
C.  $\frac{1}{3}$   
D.  $\frac{3}{16}$
- 37.** Evaluate  $2 \div \frac{1}{2}$ .  
A. 1  
B.  $\frac{1}{4}$   
C. 8  
D. 4
- 39.** Evaluate  $0.84 \div 4$ .  
A. 2.1  
B. 0.21  
C. 0.021  
D. 0.88
- 30.** Evaluate  $\frac{2}{3} \times \frac{3}{4}$ .  
A.  $\frac{6}{7}$   
B.  $\frac{8}{9}$   
C.  $\frac{1}{2}$   
D.  $\frac{5}{7}$
- 32.** Evaluate  $0.5 \times 0.2$ .  
A. 0.1  
B. 0.7  
C. 0.01  
D. 1.0
- 34.** Evaluate  $10 \div \frac{1}{4}$ .  
A. 2.5  
B. 10  
C. 40  
D. 25
- 36.** Evaluate  $\frac{1}{2} \div \frac{1}{3}$ .  
A.  $1\frac{1}{2}$   
B.  $\frac{1}{6}$   
C.  $\frac{2}{3}$   
D.  $\frac{1}{5}$
- 38.** Evaluate  $\frac{5}{6} \div 5$ .  
A.  $\frac{25}{6}$   
B.  $\frac{1}{30}$   
C.  $\frac{1}{6}$   
D.  $\frac{5}{6}$
- 40.** A sack of rice weighing 15 kg is packed into bags of  $1\frac{1}{2}$  kg each.  
How many bags are obtained?  
A. 10  
B. 15  
C. 30  
D. 5

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

(a) Benchmark fractions, percentages and decimals.

### **Question 2**

(a) Comparing and ordering.

### **Question 3**

(a) Addition and subtraction of fractions.

### **Question 4**

(a) Word problems involving addition and subtraction of fractions.

### **Question 5**

(a) Multiplying fractions and finding a fraction of a quantity.

### **Question 6**

(a) Dividing fractions.

## Chapter 13: Ch

Time Allowed: 2 Hours | Total Marks: 100

### SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. There are 60 boys and 120 girls in a school. What is the ratio of boys to girls in its simplest form?
  - A. 2 : 1
  - B. 1 : 2
  - C. 1 : 3
  - D. 180 : 60
2. Express the ratio 15 : 25 in its simplest form.
  - A. 3 : 5
  - B. 5 : 3
  - C. 1 : 2
  - D. 15 : 25
3. A box contains 8 red counters and 12 blue counters. What is the ratio of red counters to blue counters in its simplest form?
  - A. 3 : 2
  - B. 4 : 6
  - C. 1 : 2
  - D. 2 : 3
4. The ratio of mangoes to oranges in a basket is 4 : 5. If there are 20 mangoes, how many oranges are in the basket?
  - A. 16
  - B. 20
  - C. 25
  - D. 100
5. In a school the ratio of teachers to pupils is 1 : 30. If there are 4 teachers, how many pupils are there?
  - A. 34
  - B. 120
  - C. 30
  - D. 26
6. The statement "4 to 9" is written in ratio notation as
  - A. 4 : 9
  - B. 9 : 4
  - C. 4 : 5
  - D. 9 : 13
7. Simplify the ratio 24 : 36.
  - A. 12 : 18
  - B. 3 : 2
  - C. 4 : 9
  - D. 2 : 3
8. Two numbers are in the ratio 5 : 3 and their sum is 40. What are the two numbers?
  - A. 20 and 20
  - B. 30 and 10
  - C. 25 and 15
  - D. 15 and 25
9. Two quantities are in the ratio 2 : 5. If the first quantity is 14, what is the second quantity?
  - A. 35
  - B. 28
  - C. 20
  - D. 70
10. In a bag, sweets and toffees are in the ratio 7 : 3. What fraction of the total number are toffees?
  - A.  $\frac{3}{7}$
  - B.  $\frac{7}{10}$
  - C.  $\frac{7}{3}$
  - D.  $\frac{3}{10}$

- 11.** A recipe has a ratio of 3 cups of flour to 4 cups of sugar. How many cups of flour are used for each cup of sugar?
- A.  $\frac{4}{3}$  cup
  - B.  $\frac{3}{4}$  cup
  - C. 3 cups
  - D. 7 cups
- 13.** Five oranges cost GH¢12.50. What is the cost of one orange?
- A. GH¢62.50
  - B. GH¢5.00
  - C. GH¢2.50
  - D. GH¢2.05
- 15.** Ama walks 6 km in 2 hours. What is her speed as a unit rate?
- A. 12 km/h
  - B. 8 km/h
  - C. 4 km/h
  - D. 3 km/h
- 17.** Kofi types 350 words in 7 minutes. How many words does he type per minute?
- A. 343
  - B. 2,450
  - C. 50
  - D. 35
- 19.** Complete the equivalent ratio:  $5 : 7 = 20 : \square$ .
- A. 25
  - B. 30
  - C. 24
  - D. 28
- 21.** 45 sweets are shared between two children in the ratio 4 : 5. What is the smaller share?
- A. 25
  - B. 4
  - C. 20
  - D. 9
- 23.** In the sharing in Question 22, how much did Adoley, the eldest, receive?
- A. GH¢2,400.00
  - B. GH¢3,600.00
  - C. GH¢7,200.00
  - D. GH¢4,800.00
- 12.** A car travels 180 km in 3 hours. What is its speed as a unit rate?
- A. 60 km/h
  - B. 183 km/h
  - C. 540 km/h
  - D. 3 km/h
- 14.** A machine fills 240 bottles of water in 4 minutes. How many bottles does it fill per minute?
- A. 960
  - B. 60
  - C. 236
  - D. 24
- 16.** The ratio 7 : 4 can be written as the unit rate
- A.  $\frac{7}{4}$
  - B.  $\frac{4}{7}$
  - C.  $\frac{11}{4}$
  - D.  $\frac{7}{11}$
- 18.** Which of the following ratios is equivalent to 2 : 3?
- A. 3 : 2
  - B. 8 : 12
  - C. 6 : 8
  - D. 12 : 8
- 20.** Ama and Kofi share GH¢120.00 in the ratio 2 : 3. How much does Kofi receive?
- A. GH¢72.00
  - B. GH¢48.00
  - C. GH¢60.00
  - D. GH¢240.00
- 22.** Kafui, Adoley and Jantuah shared an amount of money in the ratio of their ages, 3 : 4 : 2. If Jantuah, the youngest, received GH¢2,400.00, what was the total amount shared?
- A. GH¢9,600.00
  - B. GH¢10,800.00
  - C. GH¢7,200.00
  - D. GH¢2,400.00
- 24.** In a table of equivalent ratios,  $\frac{3}{10} = \frac{6}{x}$ . What is the value of x?
- A. 20
  - B. 10
  - C. 18
  - D. 30

- 25.** The ratio of the length to the width of a rectangle is 5 : 2. If the width is 8 cm, what is the length?  
A. 10 cm  
B. 16 cm  
C. 20 cm  
D. 40 cm
- 27.** Find the number that makes the ratio 2 : 7 equivalent to  $\square$  : 35.  
A. 7  
B. 12  
C. 14  
D. 10
- 29.** The ratio of pens to pencils in a shop is 3 : 8. If there are 24 pencils, how many pens are there?  
A. 3  
B. 6  
C. 9  
D. 64
- 31.** Using the same table of equivalent ratios as in Question 30, the entry  $y \rightarrow$  GH¢40 gives the value of  $y$  as  
A. 16  
B. 21  
C. 10  
D. 12
- 33.** If  $x/6 = 10/3$ , find the value of  $x$ .  
A. 60  
B. 5  
C. 20  
D. 30
- 35.** Which of the following points lies on the graph of the ratio 2 : 5?  
A. (5, 4)  
B. (2, 10)  
C. (10, 2)  
D. (4, 10)
- 37.** If  $6 : 15 = 10 : x$ , find the value of  $x$ .  
A. 15  
B. 20  
C. 25  
D. 4
- 26.** GH¢72.00 is shared among three children in the ratio 1 : 2 : 3. What is the greatest share?  
A. GH¢24.00  
B. GH¢36.00  
C. GH¢12.00  
D. GH¢72.00
- 28.** Paint is mixed in the ratio 2 parts red to 5 parts white. How many litres of red paint are needed to make 21 litres of the mixture?  
A. 6 litres  
B. 15 litres  
C. 9 litres  
D. 2 litres
- 30.** A table of equivalent ratios is shown:  
 $3 \rightarrow$  GH¢10;  $6 \rightarrow x$ ;  $9 \rightarrow$  GH¢30. What is the value of  $x$ ?  
A. GH¢13.00  
B. GH¢20.00  
C. GH¢16.00  
D. GH¢60.00
- 32.** The table shows bags of maize and their weights: 2 bags  $\rightarrow$  100 kg; 5 bags  $\rightarrow$  250 kg; 8 bags  $\rightarrow$   $\square$  kg. What is the missing weight?  
A. 400 kg  
B. 350 kg  
C. 500 kg  
D. 300 kg
- 34.** The points (1, 3), (2, 6) and (3,  $y$ ) lie on a straight line through the origin. What is the value of  $y$ ?  
A. 12  
B. 9  
C. 6  
D. 8
- 36.** One exercise book costs GH¢12.00. What is the cost of 5 exercise books?  
A. GH¢60.00  
B. GH¢17.00  
C. GH¢72.00  
D. GH¢48.00

- 39.** On a map, 1 cm represents 5 km. Two towns are 7 cm apart on the map. What is the actual distance between the towns?
- A. 12 km
  - B. 5 km
  - C. 7 km
  - D. 35 km
- 38.** A tap flows at a constant rate and fills 45 litres of water in 9 minutes. How many litres will it fill in 15 minutes?
- A. 45 litres
  - B. 75 litres
  - C. 60 litres
  - D. 90 litres
- 40.** Three workers share GH¢450.00 in the ratio of the hours they worked, 2 : 3 : 4. What is the difference between the greatest and the least shares?
- A. GH¢100.00
  - B. GH¢50.00
  - C. GH¢200.00
  - D. GH¢250.00

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

(a) The meaning of ratio and its relationship to fractions.

### **Question 2**

(a) Unit rates.

### **Question 3**

(a) Tables of equivalent ratios and sharing in a ratio.

### **Question 4**

(a) Proportional reasoning: missing values and coordinate pairs.

### **Question 5**

(a) Mixed applications of ratio, rate and sharing.

### **Question 6**

(a) Creating and reasoning with ratios.

# Chapter 14: Ch6

Time Allowed: 2 Hours | Total Marks: 100

## SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. What is the next term of the sequence 2, 5, 8, 11, ... ?  
A. 13  
B. 12  
C. 14  
D. 15
2. What is the next term of the sequence 3, 6, 12, 24, ... ?  
A. 36  
B. 48  
C. 30  
D. 96
3. The pattern 1, 4, 9, 16, ... continues with  
A. 25  
B. 20  
C. 24  
D. 36
4. A pattern of squares is made with matchsticks. Figure 1 uses 4 sticks, Figure 2 uses 7 sticks and Figure 3 uses 10 sticks. How many sticks does Figure 4 use?  
A. 12  
B. 16  
C. 11  
D. 13
5. In the matchstick pattern in Question 4, how many more sticks does each figure use than the figure before it?  
A. 4  
B. 3  
C. 7  
D. 1
6. The letters A, B, C are written repeatedly in the order A, B, C, A, B, C, ... Which letter is written 8th?  
A. B  
B. C  
C. A  
D. AB
7. What is the next term of the sequence 100, 90, 80, 70, ... ?  
A. 65  
B. 50  
C. 60  
D. 70
8. A pattern of triangles is made with matchsticks: 1 triangle uses 3 sticks, 2 triangles use 5 sticks and 3 triangles use 7 sticks. How many sticks are needed for 5 triangles?  
A. 15  
B. 10  
C. 13  
D. 11
9. What is the next term of the sequence 1, 1, 2, 3, 5, 8, ... ?  
A. 13  
B. 11  
C. 12  
D. 16
10. Which rule generates the sequence 2, 5, 8, 11, ... ?  
A. Add 2 to each term  
B. Add 3 to each term  
C. Multiply each term by 3  
D. Subtract 3 from each term
11. "One more than twice a number" is written in symbols as  
A.  $n + 2$   
B.  $2n$   
C.  $n^2$   
D.  $2n + 1$

- 13.** The rule for a sequence is  $n \rightarrow 4n$ . What is the 5th term of the sequence?
- A. 9  
B. 45  
C. 20  
D. 40
- 15.** Which expression means “twice a number, plus 5”?
- A.  $n^2 + 5$   
B.  $2(n + 5)$   
C.  $5n + 2$   
D.  $2n + 5$
- 17.** The expression  $2n - 3$  means
- A. three times a number, minus 2  
B. twice a number, plus 3  
C. three less than twice a number  
D. two less than three times a number
- 19.** Using the rule of the machine in Question 18, what output does the input 6 give?
- A. 18  
B. 16  
C. 21  
D. 19
- 21.** What is the next term of the pattern 2, 6, 18, 54, ... ?
- A. 108  
B. 72  
C. 162  
D. 56
- 23.** A sequence starts with 3 and each next term is found by multiplying the previous term by 2 and adding 1: 3, 7, 15, 31, ... What is the next term?
- A. 47  
B. 62  
C. 32  
D. 63
- 25.** What is the 8th term of the sequence whose rule is  $3n + 1$ ?
- A. 24  
B. 11  
C. 25  
D. 27
- 12.** “One less than three times a number  $n$ ” is written in symbols as
- A.  $3n - 1$   
B.  $3 - n$   
C.  $3(n - 1)$   
D.  $n - 3$
- 14.** The rule for a sequence is  $n \rightarrow n + 7$ . What is the 10th term of the sequence?
- A. 70  
B. 17  
C. 13  
D. 10
- 16.** A rule is  $n \rightarrow n - 4$ . What is the first term of the sequence it generates?
- A. -3  
B. 3  
C. -5  
D. 5
- 18.** A function machine gives these results: input 2  $\rightarrow$  output 7; input 3  $\rightarrow$  output 10; input 4  $\rightarrow$  output 13. What is the rule of the machine?
- A.  $n + 5$   
B.  $3n + 1$   
C.  $4n - 1$   
D.  $2n + 3$
- 20.** What is the 6th term of the sequence 4, 9, 14, 19, ... ?
- A. 29  
B. 24  
C. 30  
D. 34
- 22.** What are the next two terms of the sequence 50, 45, 40, 35, ... ?
- A. 25, 20  
B. 30, 25  
C. 35, 30  
D. 20, 15
- 24.** In the matchstick pattern of Question 4, the number of sticks for Figure  $n$  is given by
- A.  $3n + 1$   
B.  $n + 3$   
C.  $3n$   
D.  $4n$

- 27.** Using the rule  $y = 3x + 2$ , what is the value of  $y$  when  $x = 10$ ?
- A. 30  
B. 23  
C. 132  
D. 32
- 29.** What are the coordinates of the origin of the number plane?
- A. (1, 1)  
B. (0, 1)  
C. (0, 0)  
D. (1, 0)
- 31.** A point has  $x$ -coordinate  $-2$  and  $y$ -coordinate  $5$ . The point is written as
- A. (2, 5)  
B. (5,  $-2$ )  
C. ( $-5$ , 2)  
D. ( $-2$ , 5)
- 33.** Using the rule  $x \rightarrow 2x + 1$ , what is the output when  $x = 3$ ?
- A. 6  
B. 5  
C. 7  
D. 9
- 35.** Which of the following points lies on the graph of  $y = 2x + 1$ ?
- A. (1, 2)  
B. (2, 3)  
C. (3, 1)  
D. (1, 3)
- 37.** The point (4,  $y$ ) lies on the graph of  $y = 2x + 1$ . What is the value of  $y$ ?
- A. 8  
B. 6  
C. 9  
D. 7
- 39.** Which of the following points has a negative  $x$ -coordinate and a positive  $y$ -coordinate?
- A. (2, 5)  
B. ( $-2$ ,  $-5$ )  
C. (5,  $-2$ )  
D. ( $-2$ , 5)
- 26.** A table of values shows  $x$ : 1, 2, 3, 4 and  $y$ : 5, 8, 11, 14. Which rule connects  $x$  and  $y$ ?
- A.  $y = 2x + 3$   
B.  $y = 3x + 2$   
C.  $y = x + 4$   
D.  $y = 4x + 1$
- 28.** The pattern 1, 8, 27, 64, ... continues with
- A. 125  
B. 100  
C. 81  
D. 216
- 30.** A point is located 3 units to the right of the origin and 4 units up. What are its coordinates?
- A. (4, 3)  
B. (3, 4)  
C. ( $-3$ , 4)  
D. (3,  $-4$ )
- 32.** The rule  $x \rightarrow 2x + 1$  is applied to the input  $x = -2$ . What is the output?
- A.  $-3$   
B.  $-4$   
C. 3  
D.  $-5$
- 34.** For the rule  $x \rightarrow 2x + 1$ , which input gives an output of 5?
- A. 5  
B. 2  
C. 3  
D. 1
- 36.** The points (0, 1), (1, 3) and (2, 5) lie on a straight line. What is the next point on the line?
- A. (3, 7)  
B. (3, 6)  
C. (4, 7)  
D. (2, 7)
- 38.** The points (1, 2), (2, 4) and (3, 6) show the cost in cedis of pencils. What is the cost of 5 pencils?
- A. GH¢5.00  
B. GH¢10.00  
C. GH¢12.00  
D. GH¢25.00

- 40.** Using the pencil-cost relation in Question 38, how many pencils can be bought for GH¢16.00?
- A. 8
  - B. 14
  - C. 16
  - D. 6

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

- (a) Extending relations and patterns.

### **Question 2**

- (a) Describing rules in words and symbols.

### **Question 3**

- (a) Identifying rules and making predictions.

### **Question 4**

- (a) Tables of values and the number plane.

### **Question 5**

- (a) Using relations to solve problems.

### **Question 6**

- (a) Creating patterns and reasoning.

## Chapter 15: Ch

Time Allowed: 2 Hours | Total Marks: 100

### SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. If  $x$  represents an unknown number, "10 more than a number  $x$ " is written in symbols as
  - A.  $x - 10$
  - B.  $10 - x$
  - C.  $x + 10$
  - D.  $10x$
2. "5 less than a certain number  $x$ " is written in symbols as
  - A.  $x - 5$
  - B.  $5 - x$
  - C.  $x + 5$
  - D.  $5x$
3. "2 more than 5 times a certain number  $x$ " is written in symbols as
  - A.  $5(x + 2)$
  - B.  $2x + 5$
  - C.  $5 + 2x$
  - D.  $5x + 2$
4. Half of a certain number  $x$  is written in symbols as
  - A.  $2x$
  - B.  $x/2$
  - C.  $x - 2$
  - D.  $x + 2$
5. When 8 times a certain number  $x$  is subtracted from 5 and the result is multiplied by 2, the expression formed is
  - A.  $2(5 - 8x)$
  - B.  $2(8x - 5)$
  - C.  $2 \times 5 - 8x$
  - D.  $(5 - 8x) \div 2$
6. Afrako is 3 years older than Maako. If Maako is now  $x$  years old, Afrako's age is
  - A.  $x - 3$
  - B.  $x + 3$
  - C.  $3x$
  - D.  $3 - x$
7. Agbolosu and Tetteh shared GH¢400.00. Tetteh had GH¢35.00 more than Agbolosu. If Agbolosu's share is GH¢ $x$ , an expression for Tetteh's share is
  - A.  $x - 35$
  - B.  $35 - x$
  - C.  $400 - x$
  - D.  $x + 35$
8. A woman buys a basket of oranges for GH¢ $x$  and sells it for GH¢ $y$ . An expression for her profit is
  - A.  $y - x$
  - B.  $x - y$
  - C.  $x + y$
  - D.  $xy$
9. Write  $x + x$  in its simplest form.
  - A.  $x^2$
  - B.  $xx$
  - C.  $2x$
  - D.  $x + x$
10. Write  $y + y + y + y$  in its simplest form.
  - A.  $y^4$
  - B.  $4y$
  - C.  $4 + y$
  - D.  $3y$
11. Simplify  $4x + 3x + x$ .
  - A.  $8x$

- B.  $7x$
- C.  $8x^2$
- D.  $12x$

**13.** Simplify  $3x - 4x - 2x$ .

- A.  $5x$
- B.  $-5x$
- C.  $3x$
- D.  $-3x$

**12.** Simplify  $3abc + 4abc + 2abc$ .

- A.  $24abc$
- B.  $a^3b^3c^3$
- C.  $9abc$
- D.  $6abc$

- 15.** Simplify  $2p - 3q + 3p + 5q$ .  
A.  $5p + 2q$   
B.  $5p - 2q$   
C.  $5p + 8q$   
D.  $6p - q$
- 17.** Simplify  $4p \times 8p^2$ .  
A.  $32p^2$   
B.  $32p^3$   
C.  $12p^3$   
D.  $32p^6$
- 19.** Simplify  $-2a \times 4a \times 5a$ .  
A.  $40a^3$   
B.  $-40a^2$   
C.  $-40a^3$   
D.  $-11a^3$
- 21.** Simplify  $12x^3y^2 \div 16xy^4$ .  
A.  $3x^2y^2/4$   
B.  $3x^2/(4y^2)$   
C.  $4y^2/(3x^2)$   
D.  $3/(4x^2y^2)$
- 23.** Simplify  $8xyz \div 16xy$ .  
A.  $2z$   
B.  $z$   
C.  $z/2$   
D.  $xyz/2$
- 25.** If  $x = 2$  and  $y = 4$ , the value of  $3xy \times 5y$  is  
A. 120  
B. 480  
C. 240  
D. 960
- 27.** If  $p = 3$  and  $z = -1$ , the value of  $4p \times 8z^2$  is  
A. -96  
B. 32  
C. -32  
D. 96
- 29.** If  $x = 5$ , the value of  $2(x + 3)$  is  
A. 13  
B. 10  
C. 16  
D. 25
- 14.** Simplify  $5x + 4 - 9y + 3x + 2y - 7$ .  
A.  $8x - 7y + 3$   
B.  $8x - 7y - 3$   
C.  $2x - 11y - 3$   
D.  $8x + 7y - 3$
- 16.** Simplify  $7xy + 5x - 4x + 2xy - 3$ .  
A.  $9xy + x + 3$   
B.  $9xy - x - 3$   
C.  $7xy + x - 3$   
D.  $9xy + x - 3$
- 18.** Simplify  $-3xy \times 5y$ .  
A.  $-15xy^2$   
B.  $15xy^2$   
C.  $-8xy^2$   
D.  $-15x^2y$
- 20.** Simplify  $5xy^2 \times 4x^4y^3$ .  
A.  $9x^5y^5$   
B.  $20x^5y^6$   
C.  $20x^9y^6$   
D.  $20x^5y^5$
- 22.** Simplify  $18x^5y^2 \div 24x^3y^2$ .  
A.  $3x^2/4$   
B.  $3x^8/4$   
C.  $4x^2/3$   
D.  $3x/4$
- 24.** Simplify  $21x^7 \div 3x^4$ .  
A.  $18x^3$   
B.  $7x^{11}$   
C.  $7x^4$   
D.  $7x^3$
- 26.** If  $x = 2$  and  $y = 4$ , simplify  $7xy + 5x - 4x + 2xy - 3$  and find its value.  
A. 71  
B. 75  
C. 67  
D. 79
- 28.** If  $x = 2$  and  $y = 4$ , the value of  $5x + 4 - 9y + 3x + 2y - 7$  is  
A. 15  
B. -15  
C. -13  
D. -41

- 31.** If  $m = 7$  and  $n = 5$ , the value of  $(m + n) \div 2$  is  
A. 12  
B. 9.5  
C. 6  
D. 1
- 33.** Simplify  $3x^2y + 2xy^2 - 4x^2y - 6xy^2$ .  
A.  $-x^2y - 4xy^2$   
B.  $x^2y - 4xy^2$   
C.  $-x^2y + 4xy^2$   
D.  $x^2y + 4xy^2$
- 35.** Simplify  $(7x^2 + 2x^2) \div 3x^2$ .  
A.  $3x^2$   
B.  $3x$   
C. 3  
D.  $9x^2$
- 37.** The statement  $3(x + 4) = 3x + 12$  illustrates the  
A. commutative property  
B. associative property  
C. distributive property  
D. identity property
- 39.** Simplify  $10x \div 2 + 3x - x$ .  
A.  $7x$   
B.  $5x$   
C.  $8x$   
D.  $12x$
- 30.** If  $a = 3$  and  $b = -2$ , the value of  $a^2 + b$  is  
A. 7  
B. 11  
C. 1  
D.  $-7$
- 32.** If  $x = -2$  and  $y = 3$ , the value of  $3x - 2y$  is  
A. 0  
B.  $-12$   
C. 12  
D.  $-6$
- 34.** Simplify  $(7x + 4x - 2x) \div 3x$ .  
A.  $3x$   
B. 9  
C.  $3x^2$   
D. 3
- 36.** Simplify  $2(3x + 4) - x$ .  
A.  $5x + 4$   
B.  $6x + 8$   
C.  $4x + 8$   
D.  $5x + 8$
- 38.** Simplify  $4(2a - 3) + 5a$ .  
A.  $13a - 3$   
B.  $13a - 12$   
C.  $20a - 3$   
D.  $9a - 12$
- 40.** Simplify  $(6m + 9) \div 3$ .  
A.  $2m + 9$   
B.  $6m + 3$   
C.  $3m + 3$   
D.  $2m + 3$

## SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

### Question 1

(a) Forming algebraic expressions.

### Question 2

(a) Adding and subtracting algebraic expressions.

### Question 3

(a) Multiplying and dividing algebraic expressions.

### Question 4

(a) Substituting values to evaluate expressions. In this question, take  $x = 2$ ,  $y = 4$ ,  $p = 3$  and  $z = -1$ .

### Question 5

(a) Using properties of operations to simplify expressions.

### Question 6

(a) Using algebraic expressions in real-life situations.

# Chapter 16: Ch8

Time Allowed: 2 Hours | Total Marks: 100

## SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. "Think of a number, double it and subtract 7. The result is 41." If the number is  $x$ , the equation formed is
  - A.  $2(x - 7) = 41$
  - B.  $7 - 2x = 41$
  - C.  $2x + 7 = 41$
  - D.  $2x - 7 = 41$
3. Adaako and Afrakoma shared 40 oranges. Afrakoma had 6 more than Adaako. If Adaako's share is  $x$ , the equation is
  - A.  $x + (x + 6) = 40$
  - B.  $x + 6 = 40$
  - C.  $x + (x - 6) = 40$
  - D.  $6x + x = 40$
5. Which word problem matches the equation  $2x - 4 = 12$ ?
  - A. When 2 is taken away from 4 times a number, the result is 12
  - B. When 4 is taken away from 2 times a certain number, the result is 12
  - C. When 4 is added to 2 times a number, the result is 12
  - D. Twice a number, minus 12, gives 4
7. Kojo had some mangoes. After giving away 5, he had 12 left. If he started with  $x$  mangoes, the equation is
  - A.  $x + 5 = 12$
  - B.  $5 - x = 12$
  - C.  $x - 5 = 12$
  - D.  $12 - x = 5$
9. The perimeter of a rectangle is 30 cm and its length is twice its width  $w$ . The equation for the perimeter is
  - A.  $w + 2w = 30$
  - B.  $2(w + 2w) = 30$
  - C.  $2w + 2 = 30$
2. The sum of the ages of two friends is 25, and the older one is 4 times the age of the younger one. If the younger one's age is  $x$ , the equation is
  - A.  $x + 4 = 25$
  - B.  $4x + x = 25$
  - C.  $4x - x = 25$
  - D.  $x + 25 = 4x$
4. Which word statement matches the equation  $x + x = 15$ ?
  - A. The difference of two equal numbers is 15
  - B. Twice a number is 30
  - C. The sum of two equal numbers is 15
  - D. A number and 15 add up to twice the number
6. "Two-thirds of a certain number is 4" is written as the equation
  - A.  $(2/3)x = 4$
  - B.  $(3/2)x = 4$
  - C.  $2x + 3 = 4$
  - D.  $x + 2/3 = 4$
8. Ama thinks of a number, adds 8 to it and multiplies the result by 3. Her answer is 45. If the number is  $x$ , the equation is
  - A.  $3x + 8 = 45$
  - B.  $x + 24 = 45$
  - C.  $8(x + 3) = 45$
  - D.  $3(x + 8) = 45$

D.  $w \times 2w = 30$

- 10.** The equation  $x + 7 = 20$  means
- A. when 7 is added to a certain number, the result is 20
  - B. seven more than 20 is the number
  - C. a number minus 7 gives 20
  - D. when 20 is added to a number,

the result is 7

- 11.** On a balance, a bag of rice of mass  $x$  kg together with a 3 kg weight balances a 7 kg weight. The equation is
- A.  $x + 7 = 3$
  - B.  $x + 3 = 7$
  - C.  $3x = 7$
  - D.  $x - 3 = 7$
- 13.** To solve  $x + 3 = 7$  using the balance method, we
- A. subtract 3 from both sides
  - B. add 3 to both sides
  - C. multiply both sides by 3
  - D. divide both sides by 3
- 15.** Using the balance idea, the solution of  $2x + 1 = 5$  is
- A. 3
  - B. 4
  - C. 1
  - D. 2
- 17.** The solution of  $3y + 4 = 2y + 8$  is
- A. 8
  - B. 4
  - C. 12
  - D. 2
- 19.** In a counter model, 3 identical cups each holding  $x$  counters and 2 loose counters balance 11 counters. The equation is
- A.  $3x + 2 = 11$
  - B.  $x + 5 = 11$
  - C.  $3(x + 2) = 11$
  - D.  $3x - 2 = 11$
- 21.** In algebra tiles, a small square unit tile represents
- A.  $x$
  - B.  $x^2$
  - C. 1
  - D. 0
- 23.** A tile model shows 2 rectangular tiles and 3 unit tiles on the left side, balancing 9 unit tiles on the right side. The equation modelled is
- A.  $2 + 3x = 9$
  - B.  $5x = 9$
- 12.** The solution of the equation  $x + 3 = 7$  from the balance in Question 11 is
- A. 3
  - B. 10
  - C. 21
  - D. 4
- 14.** On a balance, 2 identical balls and a 1 kg weight balance a 5 kg weight. If each ball has mass  $x$  kg, the equation is
- A.  $x + 2 = 5$
  - B.  $2x - 1 = 5$
  - C.  $2x + 1 = 5$
  - D.  $2(x + 1) = 5$
- 16.** To solve  $3y + 4 = 2y + 8$  using the balance method, the first step of removing  $2y$  from both sides gives
- A.  $y + 4 = 8$
  - B.  $y = 12$
  - C.  $5y + 4 = 8$
  - D.  $3y = 2y + 4$
- 18.** In a shape balance, one rhombus has mass 12 g. One pentagon and one rhombus together balance a 20 g weight. The mass of the pentagon is
- A. 32 g
  - B. 12 g
  - C. 8 g
  - D. 240 g
- 20.** When solving an equation using a balance, whatever you do to one side you must
- A. do nothing to the other side
  - B. do the opposite on the same side
  - C. undo on the same side only
  - D. do the same to the other side
- 22.** In algebra tiles, a rectangular tile represents
- A.  $x$
  - B. 1
  - C.  $x^2$
  - D.  $2x$

C.  $2x - 3 = 9$

D.  $2x + 3 = 9$

- 25.** A zero pair in algebra tiles is
- A. two negative unit tiles
  - B. one positive unit tile only
  - C. one positive tile and one negative tile of the same type
  - D. one rectangular tile and one square tile
- 27.** A tile model shows 3 rectangular tiles on the left balancing 2 rectangular tiles and 5 unit tiles on the right. The solution of the equation is
- A. 2
  - B. 5
  - C. 3
  - D. 10
- 29.** Solve  $x + 7 = 20$ .
- A. 27
  - B. 12
  - C. 13
  - D. 14
- 31.** Solve  $6z + 4 = 28$ .
- A. 8
  - B. 4
  - C. 24
  - D. 6
- 33.** Solve  $4x + 1 = 3x + 7$ .
- A. 8
  - B. 5
  - C. 2
  - D. 6
- 35.** Solve  $20 - 3k = k + 12$ .
- A. 8
  - B. 2
  - C. 4
  - D. -2
- 37.** Solve  $5x - 4 = 3x + 10$ .
- A. 7
  - B. 3
  - C. 2
  - D. 5
- 24.** The solution of the equation modelled in Question 23 is
- A. 2
  - B. 3
  - C. 6
  - D. 4
- 26.** To solve  $x + 4 = 9$  using algebra tiles, we
- A. add 4 negative unit tiles to one side only
  - B. remove 9 unit tiles from both sides
  - C. add 4 unit tiles to both sides
  - D. remove 4 unit tiles from both sides
- 28.** Using tiles, the first step in solving  $2x + 5 = x + 8$  is to
- A. remove 1 x-tile from each side
  - B. remove 5 unit tiles from each side
  - C. add 3 unit tiles to each side
  - D. remove all the tiles
- 30.** Solve  $x - 9 = 15$ .
- A. 24
  - B. 6
  - C. -6
  - D. 135
- 32.** Solve  $3x + 5 = 20$ .
- A. 15
  - B. 8
  - C. 5
  - D. 4
- 34.** Solve  $7w + 3 = 2w + 18$ .
- A. 3
  - B. 5
  - C. 15
  - D. 4
- 36.** Solve  $2x - 7 = 41$ .
- A. 17
  - B. 48
  - C. 24
  - D. 34
- 38.** In 6 years' time Kofi will be 20 years old. If his present age is  $x$ , solve  $x + 6 = 20$  to find his present age.
- A. 26 years
  - B. 14 years
  - C. 120 years

D. 8 years

**40.** A number multiplied by 5, then 3 added, gives 33. The number is

A. 7.2

B. 36

C. 6

D. 15

**39.** Solve  $4x = 48$ .

A. 44

B. 52

C. 192

D. 12

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

- (a) Translating word problems and equations.

### **Question 2**

- (a) Modelling equations on a balance.

### **Question 3**

- (a) Modelling equations with algebra tiles.

### **Question 4**

- (a) Solving linear equations.

### **Question 5**

- (a) Solving problems with linear equations.

### **Question 6**

- (a) Creating equations and reasoning.

# Chapter 17: Ch

Time Allowed: 2 Hours | Total Marks: 100

## SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. An angle of  $90^\circ$  is called
  - A. a right angle
  - B. an acute angle
  - C. an obtuse angle
  - D. a reflex angle
2. An angle of  $37^\circ$  is classified as
  - A. a right angle
  - B. an obtuse angle
  - C. an acute angle
  - D. a reflex angle
3. An angle of  $129^\circ$  is classified as
  - A. an acute angle
  - B. an obtuse angle
  - C. a right angle
  - D. a straight angle
4. An angle of  $240^\circ$  is classified as
  - A. an obtuse angle
  - B. a right angle
  - C. an acute angle
  - D. a reflex angle
5. Which of the following is an obtuse angle?
  - A.  $100^\circ$
  - B.  $90^\circ$
  - C.  $45^\circ$
  - D.  $180^\circ$
6. A reflex angle can measure
  - A.  $85^\circ$
  - B.  $200^\circ$
  - C.  $90^\circ$
  - D.  $175^\circ$
7. The angle turned through by the minute hand of a clock in 15 minutes is
  - A.  $45^\circ$
  - B.  $180^\circ$
  - C.  $30^\circ$
  - D.  $90^\circ$
8. A protractor is an instrument used for
  - A. measuring angles
  - B. measuring mass
  - C. measuring length
  - D. measuring time
9. The angles  $30^\circ$ ,  $90^\circ$ ,  $150^\circ$  and  $270^\circ$  are classified, in order, as
  - A. acute, right, reflex, obtuse
  - B. right, acute, obtuse, reflex
  - C. acute, right, obtuse, reflex
  - D. acute, obtuse, right, reflex
10. A straight angle measures
  - A.  $90^\circ$
  - B.  $180^\circ$
  - C.  $360^\circ$
  - D.  $270^\circ$
11. Which of the following is NOT an acute angle?
  - A.  $30^\circ$
  - B.  $89^\circ$
  - C.  $45^\circ$
  - D.  $91^\circ$
12. One complete turn measures
  - A.  $360^\circ$
  - B.  $180^\circ$
  - C.  $270^\circ$
  - D.  $90^\circ$
13. Two angles are complementary. If one of them is  $35^\circ$ , the other is

- A.  $145^\circ$
- B.  $45^\circ$
- C.  $55^\circ$
- D.  $65^\circ$

**15.** Complementary angles are two angles whose sum is

- A.  $180^\circ$
- B.  $360^\circ$
- C.  $45^\circ$
- D.  $90^\circ$

**14.** Two angles are supplementary. If one of them is  $110^\circ$ , the other is

- A.  $80^\circ$
- B.  $70^\circ$
- C.  $90^\circ$
- D.  $290^\circ$

- 17.** Two complementary angles are equal. Each angle measures
- A.  $90^\circ$
  - B.  $45^\circ$
  - C.  $60^\circ$
  - D.  $30^\circ$
- 19.** An angle and an angle  $20^\circ$  bigger than it are complementary. The smaller angle is
- A.  $35^\circ$
  - B.  $55^\circ$
  - C.  $25^\circ$
  - D.  $70^\circ$
- 21.** An angle and its complement are in the ratio 2 : 3. The angle is
- A.  $54^\circ$
  - B.  $36^\circ$
  - C.  $24^\circ$
  - D.  $45^\circ$
- 23.** The angles  $x^\circ$  and  $(x + 40)^\circ$  are supplementary. The value of  $x$  is
- A. 70
  - B. 110
  - C. 50
  - D. 140
- 25.** Which pair of angles is supplementary?
- A.  $45^\circ$  and  $55^\circ$
  - B.  $95^\circ$  and  $85^\circ$
  - C.  $120^\circ$  and  $70^\circ$
  - D.  $60^\circ$  and  $20^\circ$
- 27.** When two straight lines intersect, the vertically opposite angles are
- A. complementary
  - B. supplementary
  - C. unequal
  - D. equal
- 29.** Two straight lines intersect. One of the angles formed is  $65^\circ$ . Each of the two angles adjacent to it on a straight line measures
- A.  $115^\circ$
  - B.  $25^\circ$
  - C.  $125^\circ$
  - D.  $65^\circ$
- 16.** Supplementary angles are two angles whose sum is
- A.  $180^\circ$
  - B.  $90^\circ$
  - C.  $360^\circ$
  - D.  $270^\circ$
- 18.** Two supplementary angles are in the ratio 1 : 2. The smaller angle is
- A.  $90^\circ$
  - B.  $45^\circ$
  - C.  $30^\circ$
  - D.  $60^\circ$
- 20.** The supplement of an angle of  $62^\circ$  is
- A.  $28^\circ$
  - B.  $38^\circ$
  - C.  $118^\circ$
  - D.  $298^\circ$
- 22.** Three angles on a straight line are  $x^\circ$ ,  $2x^\circ$  and  $3x^\circ$ . The value of  $x$  is
- A. 60
  - B. 45
  - C. 30
  - D. 20
- 24.** Which pair of angles is complementary?
- A.  $60^\circ$  and  $40^\circ$
  - B.  $100^\circ$  and  $80^\circ$
  - C.  $30^\circ$  and  $50^\circ$
  - D.  $25^\circ$  and  $65^\circ$
- 26.** The supplement of a right angle is
- A.  $0^\circ$
  - B.  $180^\circ$
  - C.  $90^\circ$
  - D.  $45^\circ$
- 28.** Two straight lines intersect. One of the angles formed is  $50^\circ$ . The angle vertically opposite to it is
- A.  $40^\circ$
  - B.  $50^\circ$
  - C.  $130^\circ$
  - D.  $310^\circ$
- 30.** Two straight lines intersect. One angle is  $3x^\circ$  and the angle vertically opposite to it is  $(x + 80)^\circ$ . The value of  $x$  is
- A. 20

- B. 80
- C. 40
- D. 120

**31.** Adjacent angles are angles which

- A. are always equal
  - B. always add up to  $180^\circ$
  - C. are formed only by parallel lines
  - D. share a common vertex and a common arm
- 33.** Three angles at a point on one side of a straight line are  $45^\circ$ ,  $60^\circ$  and  $x^\circ$ . The value of  $x$  is
- A. 85
  - B. 105
  - C. 75
  - D. 285
- 35.** The angles around a point are  $90^\circ$ ,  $120^\circ$ ,  $60^\circ$  and  $x^\circ$ . The value of  $x$  is
- A. 30
  - B. 60
  - C. 120
  - D. 90
- 37.** The angle vertically opposite to an angle of  $142^\circ$  is
- A.  $38^\circ$
  - B.  $142^\circ$
  - C.  $48^\circ$
  - D.  $218^\circ$
- 39.** The angles  $70^\circ$  and  $110^\circ$  are best classified as a pair of
- A. complementary angles
  - B. vertically opposite angles
  - C. supplementary angles
  - D. adjacent angles
- 32.** Two angles lie on a straight line. One of them is  $70^\circ$ . The other is
- A.  $110^\circ$
  - B.  $20^\circ$
  - C.  $290^\circ$
  - D.  $90^\circ$
- 34.** Two straight lines intersect at a point. One of the four angles formed is 3 times an adjacent angle. The smaller angle is
- A.  $30^\circ$
  - B.  $45^\circ$
  - C.  $60^\circ$
  - D.  $135^\circ$
- 36.** Two adjacent angles on a straight line are such that one is twice the other. The smaller angle is
- A.  $60^\circ$
  - B.  $120^\circ$
  - C.  $45^\circ$
  - D.  $90^\circ$
- 38.** Two straight lines intersect and one of the angles formed is  $90^\circ$ . The other three angles measure
- A.  $90^\circ$ ,  $45^\circ$  and  $45^\circ$
  - B.  $180^\circ$ ,  $90^\circ$  and  $0^\circ$
  - C.  $60^\circ$ ,  $120^\circ$  and  $90^\circ$
  - D.  $90^\circ$ ,  $90^\circ$  and  $90^\circ$
- 40.** Two angles at a point on a straight edge are  $x^\circ$  and  $(x + 20)^\circ$ . The value of  $x$  is
- A. 80
  - B. 100
  - C. 70
  - D. 60

## **SECTION B: STRUCTURED QUESTIONS**

Answer all questions and show all your workings clearly.

### **Question 1**

(a) Measuring and classifying angles.

### **Question 2**

(a) Complementary and supplementary angles.

### **Question 3**

(a) Angles on straight lines and at a point.

### **Question 4**

(a) Vertically opposite angles.

### **Question 5**

(a) Mixed angle problems.

### **Question 6**

(a) Drawing angles and reasoning.

# **ANSWER KEY**

## Ch10

## ANSWER KEY

## SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                            | Cognitive Level |
|----|-----|--------------------------------------------|-----------------|
| 1  | B   | Perpendiculars & copying line segments     | Remember        |
| 2  | A   | Perpendiculars & copying line segments     | Remember        |
| 3  | C   | Perpendiculars & copying line segments     | Remember        |
| 4  | D   | Perpendiculars & copying line segments     | Understand      |
| 5  | B   | Perpendiculars & copying line segments     | Remember        |
| 6  | D   | Perpendiculars & copying line segments     | Apply           |
| 7  | A   | Perpendiculars & copying line segments     | Apply           |
| 8  | C   | Perpendiculars & copying line segments     | Understand      |
| 9  | B   | Perpendicular bisectors                    | Remember        |
| 10 | D   | Perpendicular bisectors                    | Apply           |
| 11 | A   | Perpendicular bisectors                    | Understand      |
| 12 | B   | Perpendicular bisectors                    | Apply           |
| 13 | D   | Perpendicular bisectors                    | Understand      |
| 14 | A   | Perpendicular bisectors                    | Analyse         |
| 15 | C   | Perpendicular bisectors                    | Apply           |
| 16 | B   | Perpendicular bisectors                    | Understand      |
| 17 | D   | Copying & bisecting angles; special angles | Remember        |
| 18 | A   | Copying & bisecting angles; special angles | Apply           |
| 19 | C   | Copying & bisecting angles; special angles | Apply           |
| 20 | A   | Copying & bisecting angles; special angles | Apply           |
| 21 | C   | Copying & bisecting angles; special angles | Apply           |
| 22 | A   | Copying & bisecting angles; special angles | Understand      |
| 23 | B   | Copying & bisecting angles; special angles | Apply           |
| 24 | D   | Copying & bisecting angles; special angles | Apply           |
| 25 | C   | Copying & bisecting angles; special angles | Apply           |
| 26 | D   | Copying & bisecting angles; special angles | Understand      |
| 27 | B   | Copying & bisecting angles; special angles | Apply           |
| 28 | C   | Copying & bisecting angles; special angles | Apply           |
| 29 | A   | Copying & bisecting angles; special angles | Understand      |
| 30 | D   | Copying & bisecting angles; special angles | Remember        |
| 31 | A   | Copying & bisecting angles; special angles | Remember        |
| 32 | C   | Copying & bisecting angles; special angles | Apply           |
| 33 | B   | Constructions in the environment           | Apply           |
| 34 | D   | Constructions in the environment           | Apply           |
| 35 | A   | Constructions in the environment           | Understand      |
| 36 | B   | Constructions in the environment           | Apply           |
| 37 | D   | Constructions in the environment           | Understand      |

| Q  | Ans      | Curriculum Area                  | Cognitive Level |
|----|----------|----------------------------------|-----------------|
| 38 | <b>A</b> | Constructions in the environment | Understand      |
| 39 | <b>C</b> | Constructions in the environment | Apply           |
| 40 | <b>B</b> | Constructions in the environment | Understand      |

**Answer Distribution:**

A: 11 (27%) | B: 10 (25%) | C: 9 (22%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 1 (2%) | Apply: 19 (47%) | Remember: 8 (20%) | Understand: 12 (30%)

## Ch11

## ANSWER KEY

## SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                      | Cognitive Level |
|----|-----|--------------------------------------|-----------------|
| 1  | C   | Perimeter in mixed units             | Apply           |
| 2  | B   | Perimeter in mixed units             | Apply           |
| 3  | D   | Perimeter in mixed units             | Apply           |
| 4  | A   | Perimeter in mixed units             | Apply           |
| 5  | C   | Perimeter in mixed units             | Apply           |
| 6  | A   | Perimeter in mixed units             | Apply           |
| 7  | D   | Perimeter in mixed units             | Apply           |
| 8  | B   | Perimeter in mixed units             | Apply           |
| 9  | C   | Perimeter in mixed units             | Apply           |
| 10 | A   | Perimeter in mixed units             | Apply           |
| 11 | D   | Perimeter in mixed units             | Apply           |
| 12 | B   | Perimeter in mixed units             | Analyse         |
| 13 | A   | Parts of a circle & circumference    | Remember        |
| 14 | C   | Parts of a circle & circumference    | Remember        |
| 15 | D   | Parts of a circle & circumference    | Remember        |
| 16 | B   | Parts of a circle & circumference    | Remember        |
| 17 | A   | Parts of a circle & circumference    | Understand      |
| 18 | C   | Parts of a circle & circumference    | Remember        |
| 19 | D   | Parts of a circle & circumference    | Remember        |
| 20 | B   | Parts of a circle & circumference    | Apply           |
| 21 | A   | Parts of a circle & circumference    | Apply           |
| 22 | C   | Parts of a circle & circumference    | Apply           |
| 23 | B   | Parts of a circle & circumference    | Apply           |
| 24 | D   | Parts of a circle & circumference    | Apply           |
| 25 | A   | Parts of a circle & circumference    | Apply           |
| 26 | C   | Parts of a circle & circumference    | Apply           |
| 27 | D   | Parts of a circle & circumference    | Understand      |
| 28 | A   | Parts of a circle & circumference    | Remember        |
| 29 | B   | Parts of a circle & circumference    | Understand      |
| 30 | C   | Parts of a circle & circumference    | Apply           |
| 31 | D   | Parts of a circle & circumference    | Apply           |
| 32 | B   | Parts of a circle & circumference    | Analyse         |
| 33 | D   | Grid drawing: rectangles & triangles | Remember        |
| 34 | A   | Grid drawing: rectangles & triangles | Apply           |
| 35 | C   | Grid drawing: rectangles & triangles | Apply           |
| 36 | B   | Grid drawing: rectangles & triangles | Apply           |
| 37 | A   | Grid drawing: rectangles & triangles | Remember        |

| Q  | Ans      | Curriculum Area                      | Cognitive Level |
|----|----------|--------------------------------------|-----------------|
| 38 | <b>C</b> | Grid drawing: rectangles & triangles | Apply           |
| 39 | <b>D</b> | Grid drawing: rectangles & triangles | Analyse         |
| 40 | <b>A</b> | Grid drawing: rectangles & triangles | Understand      |

**Answer Distribution:**

A: 11 (27%) | B: 9 (22%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 3 (7%) | Apply: 24 (60%) | Remember: 9 (22%) | Understand: 4 (10%)

## Ch12

## ANSWER KEY

## SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                | Cognitive Level |
|----|-----|--------------------------------|-----------------|
| 1  | A   | Deriving the area formula      | Understand      |
| 2  | D   | Deriving the area formula      | Remember        |
| 3  | B   | Deriving the area formula      | Apply           |
| 4  | C   | Deriving the area formula      | Apply           |
| 5  | A   | Deriving the area formula      | Understand      |
| 6  | C   | Deriving the area formula      | Understand      |
| 7  | B   | Deriving the area formula      | Understand      |
| 8  | D   | Deriving the area formula      | Remember        |
| 9  | A   | Deriving the area formula      | Understand      |
| 10 | C   | Deriving the area formula      | Apply           |
| 11 | D   | Deriving the area formula      | Understand      |
| 12 | A   | Deriving the area formula      | Apply           |
| 13 | C   | Calculating areas of triangles | Apply           |
| 14 | B   | Calculating areas of triangles | Apply           |
| 15 | D   | Calculating areas of triangles | Apply           |
| 16 | B   | Calculating areas of triangles | Apply           |
| 17 | C   | Calculating areas of triangles | Apply           |
| 18 | A   | Calculating areas of triangles | Apply           |
| 19 | D   | Calculating areas of triangles | Apply           |
| 20 | B   | Calculating areas of triangles | Apply           |
| 21 | A   | Calculating areas of triangles | Apply           |
| 22 | D   | Calculating areas of triangles | Apply           |
| 23 | B   | Calculating areas of triangles | Apply           |
| 24 | A   | Calculating areas of triangles | Apply           |
| 25 | C   | Calculating areas of triangles | Apply           |
| 26 | D   | Calculating areas of triangles | Analyse         |
| 27 | B   | Area problems in real life     | Apply           |
| 28 | C   | Area problems in real life     | Apply           |
| 29 | D   | Area problems in real life     | Apply           |
| 30 | A   | Area problems in real life     | Apply           |
| 31 | B   | Area problems in real life     | Apply           |
| 32 | A   | Area problems in real life     | Apply           |
| 33 | D   | Area problems in real life     | Apply           |
| 34 | C   | Area problems in real life     | Apply           |
| 35 | B   | Area problems in real life     | Analyse         |
| 36 | A   | Area problems in real life     | Apply           |
| 37 | C   | Area problems in real life     | Apply           |

| Q  | Ans      | Curriculum Area            | Cognitive Level |
|----|----------|----------------------------|-----------------|
| 38 | <b>D</b> | Area problems in real life | Apply           |
| 39 | <b>B</b> | Area problems in real life | Analyse         |
| 40 | <b>A</b> | Area problems in real life | Apply           |

**Answer Distribution:**

A: 11 (27%) | B: 10 (25%) | C: 9 (22%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 3 (7%) | Apply: 29 (72%) | Remember: 2 (5%) | Understand: 6 (15%)

## Ch13

## ANSWER KEY

## SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                | Cognitive Level |
|----|-----|--------------------------------|-----------------|
| 1  | D   | Bearings & back bearings       | Remember        |
| 2  | A   | Bearings & back bearings       | Remember        |
| 3  | C   | Bearings & back bearings       | Remember        |
| 4  | B   | Bearings & back bearings       | Apply           |
| 5  | D   | Bearings & back bearings       | Remember        |
| 6  | A   | Bearings & back bearings       | Remember        |
| 7  | B   | Bearings & back bearings       | Remember        |
| 8  | D   | Bearings & back bearings       | Apply           |
| 9  | C   | Bearings & back bearings       | Apply           |
| 10 | A   | Bearings & back bearings       | Apply           |
| 11 | C   | Bearings & back bearings       | Apply           |
| 12 | D   | Bearings & back bearings       | Remember        |
| 13 | A   | Bearings & back bearings       | Remember        |
| 14 | B   | Bearings & back bearings       | Apply           |
| 15 | C   | Bearings & back bearings       | Understand      |
| 16 | B   | Bearings & back bearings       | Apply           |
| 17 | A   | Scalars, vectors & column form | Remember        |
| 18 | C   | Scalars, vectors & column form | Understand      |
| 19 | D   | Scalars, vectors & column form | Understand      |
| 20 | B   | Scalars, vectors & column form | Understand      |
| 21 | A   | Scalars, vectors & column form | Understand      |
| 22 | C   | Scalars, vectors & column form | Understand      |
| 23 | B   | Scalars, vectors & column form | Understand      |
| 24 | D   | Scalars, vectors & column form | Understand      |
| 25 | A   | Scalars, vectors & column form | Remember        |
| 26 | D   | Scalars, vectors & column form | Understand      |
| 27 | B   | Scalars, vectors & column form | Understand      |
| 28 | C   | Scalars, vectors & column form | Understand      |
| 29 | D   | Magnitude of vectors           | Apply           |
| 30 | A   | Magnitude of vectors           | Apply           |
| 31 | B   | Magnitude of vectors           | Apply           |
| 32 | C   | Magnitude of vectors           | Apply           |
| 33 | D   | Magnitude of vectors           | Remember        |
| 34 | A   | Magnitude of vectors           | Apply           |
| 35 | B   | Magnitude of vectors           | Apply           |
| 36 | A   | Magnitude of vectors           | Apply           |
| 37 | D   | Magnitude of vectors           | Apply           |

| Q  | Ans      | Curriculum Area      | Cognitive Level |
|----|----------|----------------------|-----------------|
| 38 | <b>B</b> | Magnitude of vectors | Apply           |
| 39 | <b>C</b> | Magnitude of vectors | Apply           |
| 40 | <b>A</b> | Magnitude of vectors | Apply           |

**Answer Distribution:**

A: 11 (27%) | B: 10 (25%) | C: 9 (22%) | D: 10 (25%)

**Cognitive Level Distribution:**

Apply: 18 (45%) | Remember: 11 (27%) | Understand: 11 (27%)

# Ch14

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans      | Curriculum Area                    | Cognitive Level |
|----|----------|------------------------------------|-----------------|
| 1  | <b>B</b> | Reflectional (fold) symmetry       | Understand      |
| 2  | <b>C</b> | Reflectional (fold) symmetry       | Apply           |
| 3  | <b>A</b> | Reflectional (fold) symmetry       | Apply           |
| 4  | <b>D</b> | Reflectional (fold) symmetry       | Understand      |
| 5  | <b>B</b> | Reflectional (fold) symmetry       | Understand      |
| 6  | <b>D</b> | Reflectional (fold) symmetry       | Apply           |
| 7  | <b>A</b> | Reflectional (fold) symmetry       | Apply           |
| 8  | <b>B</b> | Reflectional (fold) symmetry       | Remember        |
| 9  | <b>C</b> | Reflection in the coordinate plane | Apply           |
| 10 | <b>D</b> | Reflection in the coordinate plane | Apply           |
| 11 | <b>A</b> | Reflection in the coordinate plane | Understand      |
| 12 | <b>C</b> | Reflection in the coordinate plane | Apply           |
| 13 | <b>D</b> | Reflection in the coordinate plane | Apply           |
| 14 | <b>B</b> | Reflection in the coordinate plane | Understand      |
| 15 | <b>A</b> | Reflection in the coordinate plane | Understand      |
| 16 | <b>C</b> | Reflection in the coordinate plane | Remember        |
| 17 | <b>B</b> | Reflection in the coordinate plane | Apply           |
| 18 | <b>A</b> | Reflection in the coordinate plane | Apply           |
| 19 | <b>D</b> | Reflection in the coordinate plane | Apply           |
| 20 | <b>C</b> | Reflection in the coordinate plane | Understand      |
| 21 | <b>D</b> | Translation by vectors             | Remember        |
| 22 | <b>A</b> | Translation by vectors             | Understand      |
| 23 | <b>C</b> | Translation by vectors             | Apply           |
| 24 | <b>B</b> | Translation by vectors             | Apply           |
| 25 | <b>D</b> | Translation by vectors             | Apply           |
| 26 | <b>A</b> | Translation by vectors             | Apply           |
| 27 | <b>B</b> | Translation by vectors             | Understand      |
| 28 | <b>D</b> | Translation by vectors             | Understand      |
| 29 | <b>A</b> | Translation by vectors             | Analyse         |
| 30 | <b>C</b> | Translation by vectors             | Analyse         |
| 31 | <b>D</b> | Congruent & similar shapes         | Remember        |
| 32 | <b>C</b> | Congruent & similar shapes         | Remember        |
| 33 | <b>A</b> | Congruent & similar shapes         | Understand      |
| 34 | <b>B</b> | Congruent & similar shapes         | Understand      |
| 35 | <b>D</b> | Congruent & similar shapes         | Apply           |
| 36 | <b>A</b> | Congruent & similar shapes         | Understand      |
| 37 | <b>D</b> | Congruent & similar shapes         | Apply           |

| Q  | Ans      | Curriculum Area            | Cognitive Level |
|----|----------|----------------------------|-----------------|
| 38 | <b>B</b> | Congruent & similar shapes | Apply           |
| 39 | <b>C</b> | Congruent & similar shapes | Apply           |
| 40 | <b>A</b> | Congruent & similar shapes | Understand      |

**Answer Distribution:**

A: 11 (27%) | B: 9 (22%) | C: 9 (22%) | D: 11 (27%)

**Cognitive Level Distribution:**

Analyse: 2 (5%) | Apply: 19 (47%) | Remember: 5 (12%) | Understand: 14 (35%)

## Ch15

## ANSWER KEY

## SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | B   | Data collection methods (quantitative & qu... | Remember        |
| 2  | C   | Data collection methods (quantitative & qu... | Understand      |
| 3  | D   | Data collection methods (quantitative & qu... | Apply           |
| 4  | A   | Data collection methods (quantitative & qu... | Apply           |
| 5  | B   | Data collection methods (quantitative & qu... | Remember        |
| 6  | A   | Data collection methods (quantitative & qu... | Remember        |
| 7  | D   | Data collection methods (quantitative & qu... | Understand      |
| 8  | C   | Data collection methods (quantitative & qu... | Apply           |
| 9  | D   | Data collection methods (quantitative & qu... | Remember        |
| 10 | B   | Data collection methods (quantitative & qu... | Remember        |
| 11 | A   | Data collection methods (quantitative & qu... | Apply           |
| 12 | C   | Data collection methods (quantitative & qu... | Understand      |
| 13 | B   | Questionnaire design & administration         | Remember        |
| 14 | C   | Questionnaire design & administration         | Understand      |
| 15 | A   | Questionnaire design & administration         | Remember        |
| 16 | B   | Questionnaire design & administration         | Understand      |
| 17 | C   | Questionnaire design & administration         | Understand      |
| 18 | B   | Questionnaire design & administration         | Remember        |
| 19 | A   | Questionnaire design & administration         | Understand      |
| 20 | C   | Questionnaire design & administration         | Understand      |
| 21 | B   | Questionnaire design & administration         | Understand      |
| 22 | A   | Questionnaire design & administration         | Apply           |
| 23 | C   | Questionnaire design & administration         | Analyse         |
| 24 | D   | Questionnaire design & administration         | Apply           |
| 25 | C   | Organising & displaying data                  | Remember        |
| 26 | D   | Organising & displaying data                  | Apply           |
| 27 | A   | Organising & displaying data                  | Apply           |
| 28 | B   | Organising & displaying data                  | Apply           |
| 29 | C   | Organising & displaying data                  | Apply           |
| 30 | A   | Organising & displaying data                  | Apply           |
| 31 | C   | Organising & displaying data                  | Understand      |
| 32 | B   | Organising & displaying data                  | Understand      |
| 33 | D   | Organising & displaying data                  | Remember        |
| 34 | A   | Organising & displaying data                  | Remember        |
| 35 | B   | Organising & displaying data                  | Apply           |
| 36 | C   | Organising & displaying data                  | Apply           |
| 37 | D   | Organising & displaying data                  | Apply           |

| Q  | Ans      | Curriculum Area              | Cognitive Level |
|----|----------|------------------------------|-----------------|
| 38 | <b>A</b> | Organising & displaying data | Understand      |
| 39 | <b>D</b> | Organising & displaying data | Understand      |
| 40 | <b>A</b> | Organising & displaying data | Understand      |

**Answer Distribution:**

A: 11 (27%) | B: 10 (25%) | C: 11 (27%) | D: 8 (20%)

**Cognitive Level Distribution:**

Analyse: 1 (2%) | Apply: 14 (35%) | Remember: 11 (27%) | Understand: 14 (35%)

**Ch16****ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

| Q  | Ans      | Curriculum Area           | Cognitive Level |
|----|----------|---------------------------|-----------------|
| 1  | <b>C</b> | The mean                  | Remember        |
| 2  | <b>A</b> | The mean                  | Apply           |
| 3  | <b>D</b> | The mean                  | Apply           |
| 4  | <b>B</b> | The mean                  | Apply           |
| 5  | <b>A</b> | The mean                  | Analyse         |
| 6  | <b>B</b> | The mean                  | Apply           |
| 7  | <b>C</b> | The mean                  | Apply           |
| 8  | <b>A</b> | The mean                  | Analyse         |
| 9  | <b>D</b> | The mean                  | Apply           |
| 10 | <b>B</b> | The mean                  | Analyse         |
| 11 | <b>D</b> | The mean                  | Apply           |
| 12 | <b>C</b> | The mean                  | Analyse         |
| 13 | <b>A</b> | The mean                  | Understand      |
| 14 | <b>B</b> | The mean                  | Understand      |
| 15 | <b>D</b> | The mean                  | Apply           |
| 16 | <b>A</b> | The mean                  | Analyse         |
| 17 | <b>B</b> | The median                | Remember        |
| 18 | <b>D</b> | The median                | Understand      |
| 19 | <b>C</b> | The median                | Apply           |
| 20 | <b>A</b> | The median                | Apply           |
| 21 | <b>B</b> | The median                | Apply           |
| 22 | <b>C</b> | The median                | Understand      |
| 23 | <b>B</b> | The median                | Apply           |
| 24 | <b>A</b> | The median                | Apply           |
| 25 | <b>C</b> | The median                | Apply           |
| 26 | <b>D</b> | The median                | Analyse         |
| 27 | <b>A</b> | The median                | Understand      |
| 28 | <b>C</b> | The median                | Analyse         |
| 29 | <b>B</b> | The mode & mixed measures | Remember        |
| 30 | <b>D</b> | The mode & mixed measures | Apply           |
| 31 | <b>C</b> | The mode & mixed measures | Apply           |
| 32 | <b>A</b> | The mode & mixed measures | Understand      |
| 33 | <b>D</b> | The mode & mixed measures | Analyse         |
| 34 | <b>B</b> | The mode & mixed measures | Understand      |
| 35 | <b>A</b> | The mode & mixed measures | Apply           |
| 36 | <b>C</b> | The mode & mixed measures | Analyse         |
| 37 | <b>D</b> | The mode & mixed measures | Understand      |

| Q  | Ans      | Curriculum Area           | Cognitive Level |
|----|----------|---------------------------|-----------------|
| 38 | <b>B</b> | The mode & mixed measures | Apply           |
| 39 | <b>C</b> | The mode & mixed measures | Apply           |
| 40 | <b>A</b> | The mode & mixed measures | Analyse         |

**Answer Distribution:**

A: 11 (27%) | B: 10 (25%) | C: 10 (25%) | D: 9 (22%)

**Cognitive Level Distribution:**

Analyse: 10 (25%) | Apply: 19 (47%) | Remember: 3 (7%) | Understand: 8 (20%)

**Ch17****ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

| Q  | Ans      | Curriculum Area                              | Cognitive Level |
|----|----------|----------------------------------------------|-----------------|
| 1  | <b>B</b> | Likelihood (impossible / possible / certain) | Remember        |
| 2  | <b>C</b> | Likelihood (impossible / possible / certain) | Apply           |
| 3  | <b>A</b> | Likelihood (impossible / possible / certain) | Apply           |
| 4  | <b>B</b> | Likelihood (impossible / possible / certain) | Understand      |
| 5  | <b>C</b> | Likelihood (impossible / possible / certain) | Remember        |
| 6  | <b>A</b> | Likelihood (impossible / possible / certain) | Remember        |
| 7  | <b>C</b> | Likelihood (impossible / possible / certain) | Apply           |
| 8  | <b>B</b> | Likelihood (impossible / possible / certain) | Understand      |
| 9  | <b>C</b> | Likelihood (impossible / possible / certain) | Apply           |
| 10 | <b>A</b> | Likelihood (impossible / possible / certain) | Apply           |
| 11 | <b>C</b> | Likelihood (impossible / possible / certain) | Apply           |
| 12 | <b>B</b> | Likelihood (impossible / possible / certain) | Apply           |
| 13 | <b>C</b> | Sample space                                 | Remember        |
| 14 | <b>B</b> | Sample space                                 | Apply           |
| 15 | <b>A</b> | Sample space                                 | Remember        |
| 16 | <b>C</b> | Sample space                                 | Understand      |
| 17 | <b>D</b> | Sample space                                 | Apply           |
| 18 | <b>A</b> | Sample space                                 | Apply           |
| 19 | <b>B</b> | Sample space                                 | Understand      |
| 20 | <b>D</b> | Sample space                                 | Remember        |
| 21 | <b>B</b> | Sample space                                 | Apply           |
| 22 | <b>C</b> | Sample space                                 | Understand      |
| 23 | <b>B</b> | Calculating probability                      | Remember        |
| 24 | <b>A</b> | Calculating probability                      | Apply           |
| 25 | <b>D</b> | Calculating probability                      | Apply           |
| 26 | <b>C</b> | Calculating probability                      | Apply           |
| 27 | <b>D</b> | Calculating probability                      | Apply           |
| 28 | <b>B</b> | Calculating probability                      | Apply           |
| 29 | <b>A</b> | Calculating probability                      | Analyse         |
| 30 | <b>D</b> | Calculating probability                      | Apply           |
| 31 | <b>A</b> | Calculating probability                      | Apply           |
| 32 | <b>B</b> | Calculating probability                      | Remember        |
| 33 | <b>C</b> | Calculating probability                      | Remember        |
| 34 | <b>D</b> | Calculating probability                      | Apply           |
| 35 | <b>A</b> | Calculating probability                      | Apply           |
| 36 | <b>D</b> | Calculating probability                      | Apply           |
| 37 | <b>A</b> | Calculating probability                      | Apply           |

| Q  | Ans      | Curriculum Area         | Cognitive Level |
|----|----------|-------------------------|-----------------|
| 38 | <b>C</b> | Calculating probability | Apply           |
| 39 | <b>B</b> | Calculating probability | Apply           |
| 40 | <b>D</b> | Calculating probability | Analyse         |

**Answer Distribution:**

A: 10 (25%) | B: 11 (27%) | C: 11 (27%) | D: 8 (20%)

**Cognitive Level Distribution:**

Analyse: 2 (5%) | Apply: 24 (60%) | Remember: 9 (22%) | Understand: 5 (12%)

**Ch1****ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

| Q  | Ans      | Curriculum Area                        | Cognitive Level |
|----|----------|----------------------------------------|-----------------|
| 1  | <b>B</b> | Place value & writing large numbers    | Remember        |
| 2  | <b>C</b> | Place value & writing large numbers    | Understand      |
| 3  | <b>B</b> | Place value & writing large numbers    | Remember        |
| 4  | <b>C</b> | Place value & writing large numbers    | Remember        |
| 5  | <b>D</b> | Place value & writing large numbers    | Apply           |
| 6  | <b>B</b> | Place value & writing large numbers    | Understand      |
| 7  | <b>A</b> | Place value & writing large numbers    | Understand      |
| 8  | <b>C</b> | Place value & writing large numbers    | Analyse         |
| 9  | <b>A</b> | Rounding whole numbers > 1,000,000,000 | Apply           |
| 10 | <b>C</b> | Rounding whole numbers > 1,000,000,000 | Apply           |
| 11 | <b>D</b> | Rounding whole numbers > 1,000,000,000 | Apply           |
| 12 | <b>B</b> | Rounding whole numbers > 1,000,000,000 | Apply           |
| 13 | <b>A</b> | Rounding whole numbers > 1,000,000,000 | Apply           |
| 14 | <b>C</b> | Rounding whole numbers > 1,000,000,000 | Analyse         |
| 15 | <b>D</b> | Rounding decimals                      | Remember        |
| 16 | <b>A</b> | Rounding decimals                      | Apply           |
| 17 | <b>B</b> | Rounding decimals                      | Apply           |
| 18 | <b>C</b> | Rounding decimals                      | Apply           |
| 19 | <b>D</b> | Rounding whole numbers > 1,000,000,000 | Understand      |
| 20 | <b>A</b> | Rounding whole numbers > 1,000,000,000 | Apply           |
| 21 | <b>B</b> | Decimal places & significant figures   | Understand      |
| 22 | <b>D</b> | Decimal places & significant figures   | Apply           |
| 23 | <b>C</b> | Decimal places & significant figures   | Apply           |
| 24 | <b>B</b> | Decimal places & significant figures   | Apply           |
| 25 | <b>A</b> | Decimal places & significant figures   | Understand      |
| 26 | <b>D</b> | Decimal places & significant figures   | Apply           |
| 27 | <b>A</b> | Comparing & ordering numbers           | Understand      |
| 28 | <b>B</b> | Comparing & ordering numbers           | Apply           |
| 29 | <b>C</b> | Comparing & ordering numbers           | Analyse         |
| 30 | <b>D</b> | Comparing & ordering numbers           | Analyse         |
| 31 | <b>B</b> | Comparing & ordering numbers           | Remember        |
| 32 | <b>C</b> | Place value & writing large numbers    | Apply           |
| 33 | <b>D</b> | Rounding whole numbers > 1,000,000,000 | Analyse         |
| 34 | <b>B</b> | Rounding whole numbers > 1,000,000,000 | Apply           |
| 35 | <b>A</b> | Rounding whole numbers > 1,000,000,000 | Understand      |
| 36 | <b>D</b> | Decimal places & significant figures   | Apply           |
| 37 | <b>C</b> | Place value & writing large numbers    | Analyse         |

| Q  | Ans      | Curriculum Area                        | Cognitive Level |
|----|----------|----------------------------------------|-----------------|
| 38 | <b>B</b> | Place value & writing large numbers    | Understand      |
| 39 | <b>A</b> | Rounding decimals                      | Apply           |
| 40 | <b>C</b> | Rounding whole numbers > 1,000,000,000 | Evaluate        |

**Answer Distribution:**

A: 9 (22%) | B: 11 (27%) | C: 11 (27%) | D: 9 (22%)

**Cognitive Level Distribution:**

Analyse: 6 (15%) | Apply: 19 (47%) | Evaluate: 1 (2%) | Remember: 5 (12%) | Understand: 9 (22%)

## Ch2

## ANSWER KEY

## SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                          | Cognitive Level |
|----|-----|------------------------------------------|-----------------|
| 1  | B   | Multiply/divide by powers of 10          | Apply           |
| 2  | C   | Multiply/divide by powers of 10          | Apply           |
| 3  | D   | Multiply/divide by powers of 10          | Apply           |
| 4  | A   | Multiply/divide by powers of 10          | Apply           |
| 5  | C   | Multiply/divide by powers of 10          | Apply           |
| 6  | B   | Multiply/divide by powers of 10          | Understand      |
| 7  | D   | Multiply/divide by powers of 10          | Understand      |
| 8  | B   | Number properties & mental strategies    | Understand      |
| 9  | C   | Number properties & mental strategies    | Understand      |
| 10 | A   | Number properties & mental strategies    | Apply           |
| 11 | D   | Number properties & mental strategies    | Remember        |
| 12 | C   | Number properties & mental strategies    | Remember        |
| 13 | A   | Number properties & mental strategies    | Apply           |
| 14 | B   | Number properties & mental strategies    | Apply           |
| 15 | D   | Number properties & mental strategies    | Analyse         |
| 16 | C   | Mental mathematics in word problems      | Apply           |
| 17 | B   | Mental mathematics in word problems      | Understand      |
| 18 | C   | Mental mathematics in word problems      | Apply           |
| 19 | A   | Addition & subtraction (up to 4-digit)   | Apply           |
| 20 | D   | Addition & subtraction (up to 4-digit)   | Apply           |
| 21 | B   | Addition & subtraction (up to 4-digit)   | Apply           |
| 22 | D   | Addition & subtraction (up to 4-digit)   | Apply           |
| 23 | A   | Addition & subtraction (up to 4-digit)   | Analyse         |
| 24 | C   | Addition & subtraction (up to 4-digit)   | Apply           |
| 25 | B   | Multiply/divide by 1- & 2-digit numbers  | Apply           |
| 26 | A   | Multiply/divide by 1- & 2-digit numbers  | Apply           |
| 27 | D   | Multiply/divide by 1- & 2-digit numbers  | Apply           |
| 28 | C   | Multiply/divide by 1- & 2-digit numbers  | Apply           |
| 29 | B   | Multiply/divide by 1- & 2-digit numbers  | Apply           |
| 30 | A   | Multiply/divide by 1- & 2-digit numbers  | Apply           |
| 31 | D   | Multiply/divide by 1- & 2-digit numbers  | Analyse         |
| 32 | C   | Multiply/divide by 1- & 2-digit numbers  | Apply           |
| 33 | A   | Decimal story problems (four operations) | Analyse         |
| 34 | D   | Decimal story problems (four operations) | Analyse         |
| 35 | B   | Decimal story problems (four operations) | Apply           |
| 36 | A   | Decimal story problems (four operations) | Apply           |
| 37 | C   | Decimal story problems (four operations) | Apply           |

| Q  | Ans      | Curriculum Area                          | Cognitive Level |
|----|----------|------------------------------------------|-----------------|
| 38 | <b>D</b> | Decimal story problems (four operations) | Apply           |
| 39 | <b>B</b> | Decimal story problems (four operations) | Apply           |
| 40 | <b>C</b> | Decimal story problems (four operations) | Apply           |

**Answer Distribution:**

A: 9 (22%) | B: 10 (25%) | C: 11 (27%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 5 (12%) | Apply: 28 (70%) | Remember: 2 (5%) | Understand: 5 (12%)

## Ch3

## ANSWER KEY

## SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                          | Cognitive Level |
|----|-----|------------------------------------------|-----------------|
| 1  | C   | Meaning of repeated factors              | Understand      |
| 2  | D   | Meaning of repeated factors              | Understand      |
| 3  | A   | Meaning of repeated factors              | Remember        |
| 4  | C   | Meaning of repeated factors              | Apply           |
| 5  | B   | Meaning of repeated factors              | Remember        |
| 6  | D   | Meaning of repeated factors              | Understand      |
| 7  | B   | Meaning of repeated factors              | Remember        |
| 8  | C   | Expressing numbers as products of powers | Apply           |
| 9  | A   | Expressing numbers as products of powers | Apply           |
| 10 | D   | Expressing numbers as products of powers | Apply           |
| 11 | C   | Expressing numbers as products of powers | Apply           |
| 12 | B   | Expressing numbers as products of powers | Apply           |
| 13 | D   | Expressing numbers as products of powers | Remember        |
| 14 | B   | Expressing numbers as products of powers | Apply           |
| 15 | A   | Expressing numbers as products of powers | Understand      |
| 16 | B   | Zero exponent ( $a^0 = 1$ )              | Remember        |
| 17 | D   | Zero exponent ( $a^0 = 1$ )              | Understand      |
| 18 | A   | Zero exponent ( $a^0 = 1$ )              | Apply           |
| 19 | C   | Zero exponent ( $a^0 = 1$ )              | Apply           |
| 20 | B   | Zero exponent ( $a^0 = 1$ )              | Understand      |
| 21 | D   | Value of numbers in index form           | Apply           |
| 22 | B   | Value of numbers in index form           | Apply           |
| 23 | A   | Value of numbers in index form           | Apply           |
| 24 | D   | Value of numbers in index form           | Apply           |
| 25 | B   | Value of numbers in index form           | Apply           |
| 26 | A   | Value of numbers in index form           | Understand      |
| 27 | C   | Value of numbers in index form           | Apply           |
| 28 | B   | Value of numbers in index form           | Apply           |
| 29 | D   | Value of numbers in index form           | Apply           |
| 30 | A   | Value of numbers in index form           | Understand      |
| 31 | C   | Value of numbers in index form           | Analyse         |
| 32 | D   | Value of numbers in index form           | Analyse         |
| 33 | B   | HCF by product of prime factors          | Apply           |
| 34 | A   | HCF by product of prime factors          | Apply           |
| 35 | C   | HCF by product of prime factors          | Apply           |
| 36 | D   | HCF by product of prime factors          | Apply           |
| 37 | B   | HCF by product of prime factors          | Analyse         |

| Q  | Ans      | Curriculum Area                 | Cognitive Level |
|----|----------|---------------------------------|-----------------|
| 38 | <b>A</b> | HCF by product of prime factors | Apply           |
| 39 | <b>D</b> | HCF by product of prime factors | Apply           |
| 40 | <b>C</b> | HCF by product of prime factors | Analyse         |

**Answer Distribution:**

A: 9 (22%) | B: 11 (27%) | C: 9 (22%) | D: 11 (27%)

**Cognitive Level Distribution:**

Analyse: 4 (10%) | Apply: 23 (57%) | Remember: 5 (12%) | Understand: 8 (20%)

## Ch4

## ANSWER KEY

## SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | C   | Benchmark fractions as percentages & decimals | Understand      |
| 2  | B   | Benchmark fractions as percentages & decimals | Apply           |
| 3  | A   | Benchmark fractions as percentages & decimals | Apply           |
| 4  | D   | Benchmark fractions as percentages & decimals | Understand      |
| 5  | C   | Benchmark fractions as percentages & decimals | Remember        |
| 6  | A   | Benchmark fractions as percentages & decimals | Remember        |
| 7  | B   | Comparing & ordering fractions, decimals, ... | Understand      |
| 8  | D   | Comparing & ordering fractions, decimals, ... | Apply           |
| 9  | C   | Comparing & ordering fractions, decimals, ... | Understand      |
| 10 | A   | Comparing & ordering fractions, decimals, ... | Apply           |
| 11 | D   | Comparing & ordering fractions, decimals, ... | Understand      |
| 12 | B   | Comparing & ordering fractions, decimals, ... | Analyse         |
| 13 | A   | Comparing & ordering fractions, decimals, ... | Analyse         |
| 14 | C   | Comparing & ordering fractions, decimals, ... | Remember        |
| 15 | B   | Addition & subtraction of fractions           | Apply           |
| 16 | D   | Addition & subtraction of fractions           | Apply           |
| 17 | A   | Addition & subtraction of fractions           | Apply           |
| 18 | C   | Addition & subtraction of fractions           | Apply           |
| 19 | B   | Addition & subtraction of fractions           | Apply           |
| 20 | A   | Addition & subtraction of fractions           | Apply           |
| 21 | D   | Addition & subtraction of fractions           | Apply           |
| 22 | C   | Word problems (add/subtract fractions)        | Apply           |
| 23 | B   | Word problems (add/subtract fractions)        | Apply           |
| 24 | A   | Word problems (add/subtract fractions)        | Apply           |
| 25 | D   | Word problems (add/subtract fractions)        | Apply           |
| 26 | C   | Multiplying fractions; fraction of a quantity | Apply           |
| 27 | B   | Multiplying fractions; fraction of a quantity | Apply           |
| 28 | A   | Multiplying fractions; fraction of a quantity | Apply           |
| 29 | D   | Multiplying fractions; fraction of a quantity | Apply           |
| 30 | C   | Multiplying fractions; fraction of a quantity | Apply           |
| 31 | B   | Multiplying fractions; fraction of a quantity | Apply           |
| 32 | A   | Multiplying fractions; fraction of a quantity | Apply           |
| 33 | D   | Multiplying fractions; fraction of a quantity | Apply           |
| 34 | C   | Dividing fractions & quantities               | Apply           |
| 35 | B   | Dividing fractions & quantities               | Apply           |
| 36 | A   | Dividing fractions & quantities               | Apply           |
| 37 | D   | Dividing fractions & quantities               | Apply           |

| Q  | Ans      | Curriculum Area                 | Cognitive Level |
|----|----------|---------------------------------|-----------------|
| 38 | <b>C</b> | Dividing fractions & quantities | Apply           |
| 39 | <b>B</b> | Dividing fractions & quantities | Apply           |
| 40 | <b>A</b> | Dividing fractions & quantities | Apply           |

**Answer Distribution:**

A: 11 (27%) | B: 10 (25%) | C: 10 (25%) | D: 9 (22%)

**Cognitive Level Distribution:**

Analyse: 2 (5%) | Apply: 30 (75%) | Remember: 3 (7%) | Understand: 5 (12%)

## Ch5

## ANSWER KEY

## SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                          | Cognitive Level |
|----|-----|------------------------------------------|-----------------|
| 1  | B   | Ratio meaning, language & simplification | Apply           |
| 2  | A   | Ratio meaning, language & simplification | Apply           |
| 3  | D   | Ratio meaning, language & simplification | Apply           |
| 4  | C   | Ratio meaning, language & simplification | Apply           |
| 5  | B   | Ratio meaning, language & simplification | Apply           |
| 6  | A   | Ratio meaning, language & simplification | Remember        |
| 7  | D   | Ratio meaning, language & simplification | Apply           |
| 8  | C   | Ratio meaning, language & simplification | Analyse         |
| 9  | A   | Ratio meaning, language & simplification | Apply           |
| 10 | D   | Ratio meaning, language & simplification | Understand      |
| 11 | B   | Unit rates                               | Understand      |
| 12 | A   | Unit rates                               | Apply           |
| 13 | C   | Unit rates                               | Apply           |
| 14 | B   | Unit rates                               | Apply           |
| 15 | D   | Unit rates                               | Apply           |
| 16 | A   | Unit rates                               | Understand      |
| 17 | C   | Unit rates                               | Apply           |
| 18 | B   | Equivalent ratios & sharing in a ratio   | Understand      |
| 19 | D   | Equivalent ratios & sharing in a ratio   | Apply           |
| 20 | A   | Equivalent ratios & sharing in a ratio   | Apply           |
| 21 | C   | Equivalent ratios & sharing in a ratio   | Apply           |
| 22 | B   | Equivalent ratios & sharing in a ratio   | Analyse         |
| 23 | D   | Equivalent ratios & sharing in a ratio   | Analyse         |
| 24 | A   | Equivalent ratios & sharing in a ratio   | Apply           |
| 25 | C   | Equivalent ratios & sharing in a ratio   | Apply           |
| 26 | B   | Equivalent ratios & sharing in a ratio   | Apply           |
| 27 | D   | Equivalent ratios & sharing in a ratio   | Apply           |
| 28 | A   | Equivalent ratios & sharing in a ratio   | Analyse         |
| 29 | C   | Equivalent ratios & sharing in a ratio   | Apply           |
| 30 | B   | Missing values & proportional reasoning  | Apply           |
| 31 | D   | Missing values & proportional reasoning  | Apply           |
| 32 | A   | Missing values & proportional reasoning  | Apply           |
| 33 | C   | Missing values & proportional reasoning  | Apply           |
| 34 | B   | Missing values & proportional reasoning  | Analyse         |
| 35 | D   | Missing values & proportional reasoning  | Analyse         |
| 36 | A   | Missing values & proportional reasoning  | Apply           |
| 37 | C   | Missing values & proportional reasoning  | Apply           |

| Q  | Ans      | Curriculum Area                         | Cognitive Level |
|----|----------|-----------------------------------------|-----------------|
| 38 | <b>B</b> | Missing values & proportional reasoning | Apply           |
| 39 | <b>D</b> | Missing values & proportional reasoning | Apply           |
| 40 | <b>A</b> | Missing values & proportional reasoning | Analyse         |

**Answer Distribution:**

A: 11 (27%) | B: 10 (25%) | C: 9 (22%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 7 (17%) | Apply: 28 (70%) | Remember: 1 (2%) | Understand: 4 (10%)

**Ch6****ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

| <b>Q</b> | <b>Ans</b> | <b>Curriculum Area</b>                  | <b>Cognitive Level</b> |
|----------|------------|-----------------------------------------|------------------------|
| 1        | <b>C</b>   | Extending relations & patterns          | Apply                  |
| 2        | <b>B</b>   | Extending relations & patterns          | Apply                  |
| 3        | <b>A</b>   | Extending relations & patterns          | Understand             |
| 4        | <b>D</b>   | Extending relations & patterns          | Apply                  |
| 5        | <b>B</b>   | Extending relations & patterns          | Understand             |
| 6        | <b>A</b>   | Extending relations & patterns          | Apply                  |
| 7        | <b>C</b>   | Extending relations & patterns          | Apply                  |
| 8        | <b>D</b>   | Extending relations & patterns          | Apply                  |
| 9        | <b>A</b>   | Extending relations & patterns          | Analyse                |
| 10       | <b>B</b>   | Describing rules (words & symbols)      | Understand             |
| 11       | <b>D</b>   | Describing rules (words & symbols)      | Understand             |
| 12       | <b>A</b>   | Describing rules (words & symbols)      | Understand             |
| 13       | <b>C</b>   | Describing rules (words & symbols)      | Apply                  |
| 14       | <b>B</b>   | Describing rules (words & symbols)      | Apply                  |
| 15       | <b>D</b>   | Describing rules (words & symbols)      | Understand             |
| 16       | <b>A</b>   | Describing rules (words & symbols)      | Apply                  |
| 17       | <b>C</b>   | Describing rules (words & symbols)      | Understand             |
| 18       | <b>B</b>   | Identifying rules & predicting elements | Analyse                |
| 19       | <b>D</b>   | Identifying rules & predicting elements | Apply                  |
| 20       | <b>A</b>   | Identifying rules & predicting elements | Apply                  |
| 21       | <b>C</b>   | Identifying rules & predicting elements | Apply                  |
| 22       | <b>B</b>   | Identifying rules & predicting elements | Apply                  |
| 23       | <b>D</b>   | Identifying rules & predicting elements | Analyse                |
| 24       | <b>A</b>   | Identifying rules & predicting elements | Analyse                |
| 25       | <b>C</b>   | Identifying rules & predicting elements | Apply                  |
| 26       | <b>B</b>   | Identifying rules & predicting elements | Analyse                |
| 27       | <b>D</b>   | Identifying rules & predicting elements | Apply                  |
| 28       | <b>A</b>   | Identifying rules & predicting elements | Understand             |
| 29       | <b>C</b>   | Tables of values & the number plane     | Remember               |
| 30       | <b>B</b>   | Tables of values & the number plane     | Remember               |
| 31       | <b>D</b>   | Tables of values & the number plane     | Remember               |
| 32       | <b>A</b>   | Tables of values & the number plane     | Apply                  |
| 33       | <b>C</b>   | Tables of values & the number plane     | Apply                  |
| 34       | <b>B</b>   | Tables of values & the number plane     | Analyse                |
| 35       | <b>D</b>   | Tables of values & the number plane     | Analyse                |
| 36       | <b>A</b>   | Tables of values & the number plane     | Apply                  |
| 37       | <b>C</b>   | Tables of values & the number plane     | Apply                  |

| Q  | Ans      | Curriculum Area                     | Cognitive Level |
|----|----------|-------------------------------------|-----------------|
| 38 | <b>B</b> | Tables of values & the number plane | Apply           |
| 39 | <b>D</b> | Tables of values & the number plane | Understand      |
| 40 | <b>A</b> | Tables of values & the number plane | Apply           |

**Answer Distribution:**

A: 11 (27%) | B: 10 (25%) | C: 9 (22%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 7 (17%) | Apply: 21 (52%) | Remember: 3 (7%) | Understand: 9 (22%)

## Ch7

## ANSWER KEY

## SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                             | Cognitive Level |
|----|-----|---------------------------------------------|-----------------|
| 1  | C   | Forming expressions from words & situations | Remember        |
| 2  | A   | Forming expressions from words & situations | Understand      |
| 3  | D   | Forming expressions from words & situations | Understand      |
| 4  | B   | Forming expressions from words & situations | Remember        |
| 5  | A   | Forming expressions from words & situations | Analyse         |
| 6  | B   | Forming expressions from words & situations | Apply           |
| 7  | D   | Forming expressions from words & situations | Apply           |
| 8  | A   | Forming expressions from words & situations | Apply           |
| 9  | C   | Adding & subtracting expressions            | Remember        |
| 10 | B   | Adding & subtracting expressions            | Remember        |
| 11 | A   | Adding & subtracting expressions            | Apply           |
| 12 | C   | Adding & subtracting expressions            | Apply           |
| 13 | D   | Adding & subtracting expressions            | Apply           |
| 14 | B   | Adding & subtracting expressions            | Analyse         |
| 15 | A   | Adding & subtracting expressions            | Apply           |
| 16 | D   | Adding & subtracting expressions            | Analyse         |
| 17 | B   | Multiplying & dividing expressions          | Apply           |
| 18 | A   | Multiplying & dividing expressions          | Apply           |
| 19 | C   | Multiplying & dividing expressions          | Apply           |
| 20 | D   | Multiplying & dividing expressions          | Analyse         |
| 21 | B   | Multiplying & dividing expressions          | Analyse         |
| 22 | A   | Multiplying & dividing expressions          | Apply           |
| 23 | C   | Multiplying & dividing expressions          | Apply           |
| 24 | D   | Multiplying & dividing expressions          | Apply           |
| 25 | B   | Substituting & evaluating                   | Apply           |
| 26 | A   | Substituting & evaluating                   | Analyse         |
| 27 | D   | Substituting & evaluating                   | Apply           |
| 28 | B   | Substituting & evaluating                   | Analyse         |
| 29 | C   | Substituting & evaluating                   | Apply           |
| 30 | A   | Substituting & evaluating                   | Apply           |
| 31 | C   | Substituting & evaluating                   | Apply           |
| 32 | B   | Substituting & evaluating                   | Apply           |
| 33 | A   | Simplifying using properties of operations  | Analyse         |
| 34 | D   | Simplifying using properties of operations  | Analyse         |
| 35 | C   | Simplifying using properties of operations  | Apply           |
| 36 | D   | Simplifying using properties of operations  | Apply           |
| 37 | C   | Simplifying using properties of operations  | Understand      |

| Q  | Ans      | Curriculum Area                            | Cognitive Level |
|----|----------|--------------------------------------------|-----------------|
| 38 | <b>B</b> | Simplifying using properties of operations | Apply           |
| 39 | <b>A</b> | Simplifying using properties of operations | Apply           |
| 40 | <b>D</b> | Simplifying using properties of operations | Apply           |

**Answer Distribution:**

A: 11 (27%) | B: 10 (25%) | C: 9 (22%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 9 (22%) | Apply: 24 (60%) | Remember: 4 (10%) | Understand: 3 (7%)

## Ch8

## ANSWER KEY

## SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                       | Cognitive Level |
|----|-----|---------------------------------------|-----------------|
| 1  | D   | Translating word problems & equations | Understand      |
| 2  | B   | Translating word problems & equations | Apply           |
| 3  | A   | Translating word problems & equations | Apply           |
| 4  | C   | Translating word problems & equations | Understand      |
| 5  | B   | Translating word problems & equations | Understand      |
| 6  | A   | Translating word problems & equations | Understand      |
| 7  | C   | Translating word problems & equations | Apply           |
| 8  | D   | Translating word problems & equations | Analyse         |
| 9  | B   | Translating word problems & equations | Apply           |
| 10 | A   | Translating word problems & equations | Understand      |
| 11 | B   | Balance & concrete material models    | Remember        |
| 12 | D   | Balance & concrete material models    | Apply           |
| 13 | A   | Balance & concrete material models    | Understand      |
| 14 | C   | Balance & concrete material models    | Apply           |
| 15 | D   | Balance & concrete material models    | Apply           |
| 16 | A   | Balance & concrete material models    | Apply           |
| 17 | B   | Balance & concrete material models    | Apply           |
| 18 | C   | Balance & concrete material models    | Apply           |
| 19 | A   | Balance & concrete material models    | Apply           |
| 20 | D   | Balance & concrete material models    | Remember        |
| 21 | C   | Algebra tile models                   | Remember        |
| 22 | A   | Algebra tile models                   | Remember        |
| 23 | D   | Algebra tile models                   | Understand      |
| 24 | B   | Algebra tile models                   | Apply           |
| 25 | C   | Algebra tile models                   | Understand      |
| 26 | D   | Algebra tile models                   | Apply           |
| 27 | B   | Algebra tile models                   | Apply           |
| 28 | A   | Algebra tile models                   | Analyse         |
| 29 | C   | Solving linear equations              | Apply           |
| 30 | A   | Solving linear equations              | Apply           |
| 31 | B   | Solving linear equations              | Apply           |
| 32 | C   | Solving linear equations              | Apply           |
| 33 | D   | Solving linear equations              | Apply           |
| 34 | A   | Solving linear equations              | Apply           |
| 35 | B   | Solving linear equations              | Analyse         |
| 36 | C   | Solving linear equations              | Apply           |
| 37 | A   | Solving linear equations              | Apply           |

| Q  | Ans      | Curriculum Area          | Cognitive Level |
|----|----------|--------------------------|-----------------|
| 38 | <b>B</b> | Solving linear equations | Apply           |
| 39 | <b>D</b> | Solving linear equations | Apply           |
| 40 | <b>C</b> | Solving linear equations | Apply           |

**Answer Distribution:**

A: 11 (27%) | B: 10 (25%) | C: 10 (25%) | D: 9 (22%)

**Cognitive Level Distribution:**

Analyse: 3 (7%) | Apply: 25 (62%) | Remember: 4 (10%) | Understand: 8 (20%)

## Ch9

## ANSWER KEY

## SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                       | Cognitive Level |
|----|-----|---------------------------------------|-----------------|
| 1  | A   | Measuring & classifying angles        | Remember        |
| 2  | C   | Measuring & classifying angles        | Apply           |
| 3  | B   | Measuring & classifying angles        | Apply           |
| 4  | D   | Measuring & classifying angles        | Apply           |
| 5  | A   | Measuring & classifying angles        | Apply           |
| 6  | B   | Measuring & classifying angles        | Understand      |
| 7  | D   | Measuring & classifying angles        | Apply           |
| 8  | A   | Measuring & classifying angles        | Understand      |
| 9  | C   | Measuring & classifying angles        | Apply           |
| 10 | B   | Measuring & classifying angles        | Remember        |
| 11 | D   | Measuring & classifying angles        | Apply           |
| 12 | A   | Measuring & classifying angles        | Remember        |
| 13 | C   | Complementary & supplementary angles  | Apply           |
| 14 | B   | Complementary & supplementary angles  | Apply           |
| 15 | D   | Complementary & supplementary angles  | Remember        |
| 16 | A   | Complementary & supplementary angles  | Remember        |
| 17 | B   | Complementary & supplementary angles  | Apply           |
| 18 | D   | Complementary & supplementary angles  | Apply           |
| 19 | A   | Complementary & supplementary angles  | Apply           |
| 20 | C   | Complementary & supplementary angles  | Apply           |
| 21 | B   | Complementary & supplementary angles  | Analyse         |
| 22 | C   | Complementary & supplementary angles  | Apply           |
| 23 | A   | Complementary & supplementary angles  | Apply           |
| 24 | D   | Complementary & supplementary angles  | Apply           |
| 25 | B   | Complementary & supplementary angles  | Apply           |
| 26 | C   | Complementary & supplementary angles  | Understand      |
| 27 | D   | Adjacent & vertically opposite angles | Remember        |
| 28 | B   | Adjacent & vertically opposite angles | Apply           |
| 29 | A   | Adjacent & vertically opposite angles | Apply           |
| 30 | C   | Adjacent & vertically opposite angles | Analyse         |
| 31 | D   | Adjacent & vertically opposite angles | Understand      |
| 32 | A   | Adjacent & vertically opposite angles | Apply           |
| 33 | C   | Adjacent & vertically opposite angles | Apply           |
| 34 | B   | Adjacent & vertically opposite angles | Analyse         |
| 35 | D   | Adjacent & vertically opposite angles | Apply           |
| 36 | A   | Adjacent & vertically opposite angles | Analyse         |
| 37 | B   | Adjacent & vertically opposite angles | Apply           |

| Q  | Ans      | Curriculum Area                       | Cognitive Level |
|----|----------|---------------------------------------|-----------------|
| 38 | <b>D</b> | Adjacent & vertically opposite angles | Understand      |
| 39 | <b>C</b> | Adjacent & vertically opposite angles | Apply           |
| 40 | <b>A</b> | Adjacent & vertically opposite angles | Apply           |

**Answer Distribution:**

A: 11 (27%) | B: 10 (25%) | C: 9 (22%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 4 (10%) | Apply: 25 (62%) | Remember: 6 (15%) | Understand: 5 (12%)

# Mathematics Formula Sheet — Basic 7

## Numbers and Operations

|                                     |                                                                           |
|-------------------------------------|---------------------------------------------------------------------------|
| <b>Place Value</b>                  | Each digit position represents $10\times$ the position to its right       |
| <b>Addition</b>                     | $a + b = \text{sum}$                                                      |
| <b>Subtraction</b>                  | $a - b = \text{difference}$                                               |
| <b>Multiplication</b>               | $a \times b = \text{product}$                                             |
| <b>Division</b>                     | $a \div b = \text{quotient}$                                              |
| <b>Order of Operations</b>          | BODMAS: Brackets, Orders, Division, Multiplication, Addition, Subtraction |
| <b>Factors</b>                      | Numbers that divide exactly into another number                           |
| <b>Multiples</b>                    | Results of multiplying a number by whole numbers                          |
| <b>HCF (Highest Common Factor)</b>  | Largest number that divides two or more numbers                           |
| <b>LCM (Lowest Common Multiple)</b> | Smallest number that is a multiple of two or more numbers                 |

## Fractions, Decimals, and Percentages

|                               |                                                            |
|-------------------------------|------------------------------------------------------------|
| <b>Fraction</b>               | numerator / denominator                                    |
| <b>Equivalent Fractions</b>   | $a/b = (a \times n)/(b \times n)$                          |
| <b>Fraction to Decimal</b>    | $a/b = a \div b$                                           |
| <b>Decimal to Fraction</b>    | $0.x = x/10$ , $0.xy = xy/100$                             |
| <b>Fraction to Percentage</b> | $a/b \times 100\%$                                         |
| <b>Percentage to Fraction</b> | $x\% = x/100$                                              |
| <b>Percentage to Decimal</b>  | $x\% = x/100 = 0.x$                                        |
| <b>Percentage of a Number</b> | $x\% \text{ of } y = (x/100) \times y$                     |
| <b>Profit/Loss Percentage</b> | $(\text{Profit or Loss} / \text{Cost Price}) \times 100\%$ |

## Algebra

|                     |                                                        |
|---------------------|--------------------------------------------------------|
| <b>Variable</b>     | A letter representing an unknown value                 |
| <b>Expression</b>   | Combination of numbers and variables (e.g., $2x + 3$ ) |
| <b>Equation</b>     | Two expressions set equal (e.g., $2x + 3 = 7$ )        |
| <b>Solve for x</b>  | Isolate x on one side of the equation                  |
| <b>Substitution</b> | Replace variables with their values                    |

## Geometry and Measurement

|                                  |                                                                         |
|----------------------------------|-------------------------------------------------------------------------|
| <b>Perimeter of Square</b>       | $P = 4 \times \text{side}$                                              |
| <b>Perimeter of Rectangle</b>    | $P = 2(\text{length} + \text{width})$                                   |
| <b>Perimeter of Triangle</b>     | $P = a + b + c$                                                         |
| <b>Circumference of Circle</b>   | $C = \pi d = 2\pi r$                                                    |
| <b>Area of Square</b>            | $A = \text{side} \times \text{side} = \text{side}^2$                    |
| <b>Area of Rectangle</b>         | $A = \text{length} \times \text{width}$                                 |
| <b>Area of Triangle</b>          | $A = \frac{1}{2} \times \text{base} \times \text{height}$               |
| <b>Area of Circle</b>            | $A = \pi r^2$                                                           |
| <b>Area of Parallelogram</b>     | $A = \text{base} \times \text{height}$                                  |
| <b>Area of Trapezium</b>         | $A = \frac{1}{2}(a + b) \times h$                                       |
| <b>Volume of Cube</b>            | $V = \text{side} \times \text{side} \times \text{side} = \text{side}^3$ |
| <b>Volume of Cuboid</b>          | $V = \text{length} \times \text{width} \times \text{height}$            |
| <b>Angles on a Straight Line</b> | $\text{Sum} = 180^\circ$                                                |
| <b>Angles at a Point</b>         | $\text{Sum} = 360^\circ$                                                |
| <b>Angles in a Triangle</b>      | $\text{Sum} = 180^\circ$                                                |
| <b>Angles in a Quadrilateral</b> | $\text{Sum} = 360^\circ$                                                |
| <b>Complementary Angles</b>      | $\text{Sum} = 90^\circ$                                                 |
| <b>Supplementary Angles</b>      | $\text{Sum} = 180^\circ$                                                |

## Data Handling

|                       |                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Mean (Average)</b> | $\text{Mean} = \text{Sum of all values} \div \text{Number of values}$                           |
| <b>Median</b>         | Middle value when data is arranged in order                                                     |
| <b>Mode</b>           | Most frequently occurring value                                                                 |
| <b>Range</b>          | $\text{Range} = \text{Highest value} - \text{Lowest value}$                                     |
| <b>Probability</b>    | $P(\text{event}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$ |

# Curriculum Reference

This book aligns with the following curriculum standards:

- Ghana Education Service (GES)  
Standards-Based Curriculum for Junior High Schools (2019)
- National Council for Curriculum and Assessment (NaCCA)  
Curriculum Framework for Basic Education
- West African Examinations Council (WAEC)  
BECE Syllabus for Integrated Science / Mathematics

The content has been reviewed to ensure alignment with:

- Strand and sub-strand organization
- Learning indicators and objectives
- Cognitive levels (Knowledge, Understanding, Application, Analysis)
- Core competencies (Critical Thinking, Problem Solving, Communication)

Teachers and students should use this book in conjunction with the official curriculum documents available from the Ghana Education Service.

For more information:

Ghana Education Service: [www.ges.gov.gh](http://www.ges.gov.gh)

NaCCA: [www.nacca.gov.gh](http://www.nacca.gov.gh)

WAEC: [www.waec.org](http://www.waec.org)

# About Beacon Educational Consult

Beacon Educational Consult is a Ghanaian educational publishing company dedicated to producing high-quality learning materials for Junior High School students across Ghana.

Our mission is to support students in achieving academic excellence through comprehensive, curriculum-aligned resources that prepare them for the Basic Education Certificate Examination (BECE) and beyond.

We believe that every student deserves access to excellent educational materials, and we are committed to making quality textbooks affordable and accessible to schools and families throughout Ghana.

Our books are:

- Written by experienced Ghanaian educators
- Aligned with the national curriculum
- Reviewed for accuracy and quality
- Designed to support both classroom learning and independent study

For more information about our publications and educational resources:

Website: [www.beaconeducational.com](http://www.beaconeducational.com)

Email: [info@beaconeducational.com](mailto:info@beaconeducational.com)

Phone: [Contact information]

Follow us on social media:

Facebook: @BeaconEducational

Instagram: @BeaconEducational

Twitter: @BeaconEdu

[illegible]

## This image shows a full page of blank, lined paper. It features approximately 20 evenly spaced horizontal grey lines across the entire width of the page, providing a guide for writing. The background is a solid off-white color. There are no margins, text, or other markings present.